Does the Gender Ratio at Colleges Affect High School Students' College Choices?*

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Abstract

Female students may not choose STEM programs in college, even when they excel in mathematics, because these programs are male-dominated and would make them a minority. Using an incentivized discrete choice experiment with high school students, we show that the gender ratio affects both female and male students' college choices: they prefer gender-balanced programs and prefer being a majority to a minority, primarily because they anticipate difficulty fitting in as minorities. Importantly, these preferences differ by ability and field: female students with high mathematics ability show stronger minority avoidance in STEM than in non-STEM, while male students show weaker minority avoidance in STEM than in non-STEM. These preferences, together with the low female share in STEM, lead to inefficient talent allocation: students with high mathematics ability – especially female students – forgo STEM, and their seats are filled by students with low mathematics ability. We validate experimental choices against actual choices several months later. These findings suggest that the low female share in STEM deters female students from entering these fields.

JEL Classification: J16, J24, I24

Keywords: STEM Gender Gap, College Choice, Gender Ratio, Talent Allocation, Preference Elicitation, Discrete Choice Experiment

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1 Introduction

Although the gender differences in mathematics and sciences are negligible at age 15 (OECD 2019), large gender gaps persist in Science, Technology, Engineering, and Mathematics (STEM) major choices at the college level: female students are 30.8 percentage points less likely than male students to major in STEM.[1] This gap may lead to several social issues, such as inefficient allocation of talent, gender-biased research topics (Truffa and Wong 2025), and gender-biased product innovation (Einiö, Feng, and Jaravel 2025; Koning, Samila, and Ferguson 2021). It can also lead to the so-called "gender data gap," where data used to set medical and safety standards disproportionately reflects men (Perez 2019).

One potential reason that has received little attention in the literature is that STEM programs are male-dominated, making female students a minority.[2] As a result, female students may avoid such programs in anticipation of the cost of being a minority. Indeed, the literature documents various such costs, including a worse classroom environment (Gong, Lu, and Song 2021), less influence in study groups (Karpowitz et al. 2024), limited peer networking opportunities (Hampole, Truffa, and Wong 2024), and a higher likelihood of dropping out (Bostwick and Weinberg 2022).

This paper studies whether the gender ratio at colleges affects high school students' college choices and whether the low female share in STEM leads to inefficient talent allocation. Answering these questions with observational data poses two challenges: the gender ratio correlates with other program attributes, making it difficult to isolate its effect; and most STEM programs are male-dominated, leaving little variation to identify how higher female shares would affect choices.

We address these challenges using an incentivized discrete choice experiment with high school students. The experiment elicits preferences over independently varied program attributes – including the student gender ratio across its full range – thereby isolating the effect of gender composition from confounding attributes. We incentivize truthful responses using the incentivized resume rating method (Kessler, Low, and Sullivan 2019), providing students with career advice tailored to their choices. We validate experimental choices against actual choices several months later.

We find that the gender ratio affects both female and male students' college choices. Both prefer gender-balanced programs over male- or female-majority programs, and both prefer being a majority to being a minority. Relative to a gender-balanced program (50% own gender), being a minority (25% own gender) reduces choice probability by 11 percentage points for female students and 9 percentage points for male students; being a majority (75% own gender) reduces choice probability by 6.5 percentage points for both genders. A decomposition reveals that students avoid being a minority primarily because they anticipate difficulty fitting in.

We also find that preferences for the gender ratio differ between STEM and non-STEM programs in ways that matter for talent allocation. Female students in the top half of mathematics ability

1. Based on the number of bachelor program graduates in 2023 from OECD Data Explorer, "Number of enrolled students, graduates and new entrants by field of education" dataset (accessed November 6, 2025).

2. A notable exception is Ersoy and Speer (2025), who show that providing non-job-related major information to students, including student gender composition, changes their major choices.

exhibit stronger minority avoidance in STEM than in non-STEM: relative to a gender-balanced program, being a minority (25% female) reduces their choice probability by 14 percentage points in STEM versus 7 percentage points in non-STEM – a twofold difference. Male students show the opposite pattern: they are less sensitive to gender composition in STEM than in non-STEM. Being a minority (25% male) reduces male students' choice probability by 8 percentage points in STEM versus 12 percentage points in non-STEM. These preferences imply a potential inefficiency in talent allocation.

We then quantify the inefficiency in talent allocation by comparing the baseline scenario – where the female share in STEM is 25% – against a benchmark scenario of the best allocation attainable where students' preferences for the gender ratio are removed.[3] In our exercise, the low female share in STEM reduces the average STEM student's mathematics ability by 0.2 standard deviations. This drop arises because students with high mathematics ability – especially female students – are deterred from entering STEM, and their seats are filled by students with low mathematics ability. The average non-STEM student's reading ability rises, but by only one-third as much. We find similar patterns using data on actual college programs.

Finally, we examine whether students choose their preferred field, using actual choices collected several months after the experiment. Most students (77.1%) choose according to their preferences, but female students who prefer STEM are less likely to follow through than their male counterparts. Since our classification captures intrinsic field preferences but not preferences over gender composition, this gap is consistent with our main finding: female students who prefer STEM may ultimately avoid it due to anticipated difficulty fitting into a male-dominated environment.

Related Literature Our paper contributes to several strands of literature. First, we identify a novel source of the STEM gender gap in higher education: the gender composition itself. Prior work has identified numerous contributing factors, which we group into three categories. (i) *Individual factors* include male students' stronger preferences for competition (Buser, Niederle, and Oosterbeek 2014) and female students' comparative advantage in non-STEM (Breda and Napp 2019; Goulas, Griselda, and Megalokonomou 2024). (ii) *Environmental factors* include gender-biased teachers (Carlana 2019; Miserocchi 2024), parental expectations (Carlana and Corno 2024), lack of role models (Breda et al. 2023; Carrell, Page, and West 2010; Riise, Willage, and Willén 2022; Riley 2024), non-cooperative classroom dynamics (Di Tommaso et al. 2024), and insufficient emphasis on STEM's social relevance (Long and Takahashi 2025). (iii) *Anticipated returns* include preferences for job amenities (Burbano, Padilla, and Meier 2024; Zafar 2013; Wiswall and Zafar 2018) and marriage-market considerations (Wiswall and Zafar 2021). Peer effects also play a role (Bechichi and Kenedi 2024; Valdebenito 2023; Mouganie and Wang 2020; Fischer 2017).[4] Our effect size – an

3. Setting the STEM female share to 50% instead of removing gender preferences yields essentially the same results.

4. Bechichi and Kenedi (2024) show that recent graduates' college choices affect high school students' college choices, and that female students respond more strongly to female graduates' choices. Valdebenito (2023) show that female classmates' college choices affect female students' college choices. Mouganie and Wang (2020) show that having high-performing female peers in the same high school class increases female students' science-track choices, whereas having high-performing male peers decreases female students' science-track choices. Fischer (2017) finds similar results

11-percentage-point increase in female students' choice probability when moving from 25% to 50% female – is economically meaningful: it corresponds to roughly one-fourth of the baseline STEM choice probability among female students under no constraints.[5]

Our study is closest to Carlana and Corno (2025), who show that female junior high students are less likely to choose counter-stereotypical tasks when they expect male-dominated peer groups, while male students are unaffected. We extend this work in two ways. First, we elicit preferences over actual college attributes rather than experimental tasks. Second, students' responses in our experiment are private – unobserved by peers, parents, or teachers – thereby ruling out social image concerns as a driver of our results. Our paper also relates to Ersoy and Speer (2025), who show that providing gender composition information changes students' major choices. We go further by recovering preferences over the full range of female shares and identifying the mechanisms underlying these preferences.

Second, we extend the emerging literature on preferences for gender diversity from workplace to educational settings. Existing work shows that workers – especially women – prefer gender-diverse workplaces, and that this preference contributes to occupational segregation (Schuh 2024; Högn et al. 2025; Funk, Iriberri, and Savio 2024). We document analogous preferences among high school students' college choices, suggesting these preferences form before labor market entry. This implies that interventions targeting educational choices may be effective in facilitating STEM-related human capital accumulation among women, compared to those targeting workplace that may only address job sorting. Our finding that making STEM programs more gender-balanced improves the allocation of talent aligns with documented benefits of gender diversity,[6] including reduced harassment risk (Folke and Rickne 2022), better recognition of ideas and talent (Koffi 2025; Bello, Casarico, and Nozza 2025), and improved team performance (Hoogendoorn, Oosterbeek, and van Praag 2013).[7]

Finally, our findings inform policies aimed at closing the STEM gender gap. While such policies – role model exposure (Breda et al. 2023; Riise, Willage, and Willén 2022; Riley 2024),[8] mentorship (Canaan and Mouganie 2023; Carrell, Page, and West 2010), single-sex instruction (Booth, Cardona-Sosa, and Nolen 2018), and pedagogical changes (Di Tommaso et al. 2024; Long and Takahashi 2025)[9] – are often justified on equity grounds, our results provide a complementary efficiency-based rationale.[10] This may help address concerns from those who view such policies as zero-sum.[11]

as Mouganie and Wang, but in a college setting and regardless of the gender of the peers.

5. The baseline STEM choice probability among female students in our sample is approximately 42.1% under no constraints.

6. On the negative side, Shan (2024) finds that students adopt more traditional gender roles when assigned to a mixed-gender study group in college.

7. In the context of family socioeconomic status (SES), Tadjfar and Vira (2025) find that students from low-SES backgrounds are reluctant to apply for elite colleges dominated by students from high-SES backgrounds due to concerns about fitting in.

8. Porter and Serra (2020) show the effectiveness of female role models in inspiring female college students to pursue an economics major.

9. Avery et al. (2024) and Owen and Hagstrom (2021) also find similar results in economics curricula.

10. While not in education, Baltrunaite et al. (2014) and Besley et al. (2017) show that a gender quota for political candidates improved elected politicians' competence. We expect similar effects for women in STEM in higher education.

11. For example, The Economist (2024) shows that some young men in Europe consider that these policies may reduce their opportunities.

The remainder of the paper is structured as follows. Section 2 explains the institutional background. Section 3 details the experimental design. Section 4 describes the summary statistics of the experimental data. Section 5 presents the main results. Section 6 quantifies the degree of inefficient allocation of talent. Section 7 validates experimental choices against actual choices. Section 8 concludes.

2 Institutional Background

We conduct our experiment in the Greater Tokyo area in Japan, which offers an ideal setting for several reasons. First, the pathway from high school to college follows a relatively uniform institutional structure compared with Europe or the US, allowing us to define hypothetical programs using attributes that students commonly observe and understand. Second, colleges and programs are ranked by a widely recognized selectivity index, which allows us to control for perceived program prestige and expected labor-market prospects. Third, college programs are densely concentrated in the Greater Tokyo area, making the hypothetical program attributes appear realistic to students and limiting unstated geographic and financial constraints on their choices.

High School High school runs from grades 10 to 12, typically from age 15 to 18 in Japan. Though not compulsory, nearly 99% of junior high school graduates attend high school (Ministry of Education, Culture, Sports, Science and Technology 2021). College enrollment rates are also high, at nearly 60%.[12] However, colleges vary widely in academic orientation: while some are research-focused, many function more like vocational institutions. We refer to the former as selective colleges and the latter as non-selective colleges. Because STEM fields require strong mathematical preparation, research-oriented STEM programs are concentrated at selective colleges. As such, academic high schools are the primary pathway for students aiming to enter selective colleges or research-oriented STEM programs.

Students at academic high schools choose a track at the end of grade 10, which determines the subjects they study in grades 11 and 12. There are two tracks: humanities and sciences. In the humanities track, students study advanced reading (advanced Japanese), English, history/social studies, and mathematics. In the sciences track, students study reading (Japanese), English, sciences (biology, chemistry, and/or physics), and advanced mathematics. A similar track system exists in many European countries.[13]

Students in these academic high schools regularly take mock exams to prepare for college entrance exams. On the score reports for these mock exams, students can see the likelihood of admission to their preferred programs, calculated based on their exam scores and the program's selectivity,

12. As of 2023: https://www.ipss.go.jp/syoushika/tohkei/Data/Popular2024/T11-03.htm (accessed June 9, 2025).

13. In Italy, students choose their track between humanities and sciences at the end of grade 9 (see, for example, Carlana and Corno 2025). In France, students choose their track between humanities, social sciences, and sciences at the end of grade 10 (see, for example, Breda et al. 2023). In the Netherlands, students in academic secondary schools (VWO) choose their track between science, health, social sciences, and humanities at the end of grade 9 (see, for example, Buser, Niederle, and Oosterbeek 2014).

which they use to make and adjust their study plan. To help students select programs, nearly all academic high schools maintain counselor offices with information about each program, including selectivity, gender ratio, and other attributes used in this paper's experiment.

College Application As in Europe but unlike in the US (Bordon and Fu 2015), students apply to specific college programs and cannot change their majors later. Programs differ in their attributes, such as major, selectivity, tuition, whether they are public or private, and location, among others. Since living alone can be costly and some parents prefer their children not to live alone, many college students live with their parents. Among various locations in Japan, the Greater Tokyo area offers the greatest variety of college attributes: nearly 29% of all colleges in Japan are located there, where about 41% of all college students study.[14]

Most selective college programs employ an exam-based, meritocratic admissions system. They rank applicants by exam score and make offers from the top of the list. However, each program requires exams in different subjects. Humanities and social sciences programs usually require exams in advanced reading, English, history/social studies, and mathematics. On the other hand, science, engineering, and medicine programs typically require exams in advanced mathematics, English, sciences, and reading. Since students who choose the science track in high school do not study advanced reading and history/social studies, they are effectively constrained from applying to most humanities and social sciences programs. Similarly, students who choose the humanities track in high school do not study advanced mathematics and sciences, making it difficult to apply to science, engineering, and medicine programs. In this way, high school track choice largely determines the set of college majors students can realistically pursue.

Each college program is assigned a single number called the "selectivity index" by commercial college entrance exam preparation companies. The index represents the difficulty of admission and the selectivity (or prestige) of the program and is calculated based on admitted students' mock exam performance. The index is expressed as a z-score rescaled to have a mean of 50 and a standard deviation of 10. The selectivity of the program from which a student graduates significantly affects the quality of the first job they obtain (Nakajima 2018) as well as promotions in the first few years after starting that job (Araki, Kawaguchi, and Onozuka 2016). Given the rigid labor market and limited job mobility in Japan (Moriguchi 2014), the quality of the first job affects students' career prospects more than in other OECD countries (Genda, Kondo, and Ohta 2010).[15]

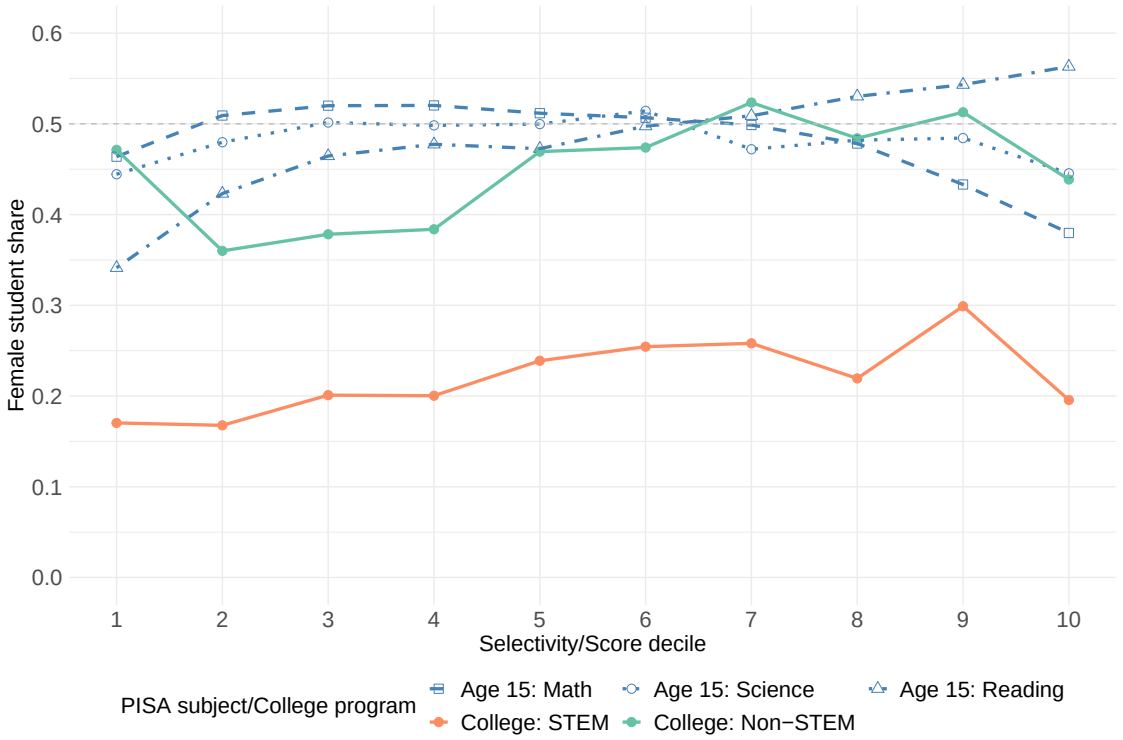
Gender Gap in STEM and Its Persistence Before describing our experiment, we document the gender gap in mathematics at age 15 and in college programs, as well as the persistence of gender ratios that our experiment aims to explain.

Figure 1 plots the share of female students in each decile of the PISA mathematics (dark dashed),

14. From an article by an educational materials and news publishing company Obunsha: https://eic.obunsha.co.jp/file/educational_info/2024/1022.pdf (accessed July 31, 2025).

15. Genda, Kondo, and Ohta (2010) find that those who entered the labor market during a recession suffer worse employment conditions in terms of unemployment and earnings in Japan than in the US.

Figure 1: Female share at age 15 and colleges by subject/program



Notes: This figure plots the share of female students in each decile of the PISA mathematics (dark dashed), science (dark dotted), and reading (dark dash-dotted) score distributions among 15-year-old students in 2018 who planned to attend a four-year college, as well as the share of female students in STEM (orange) and non-STEM (green) college programs in 2023. *Sources:* OECD (2018), School Basic Survey.

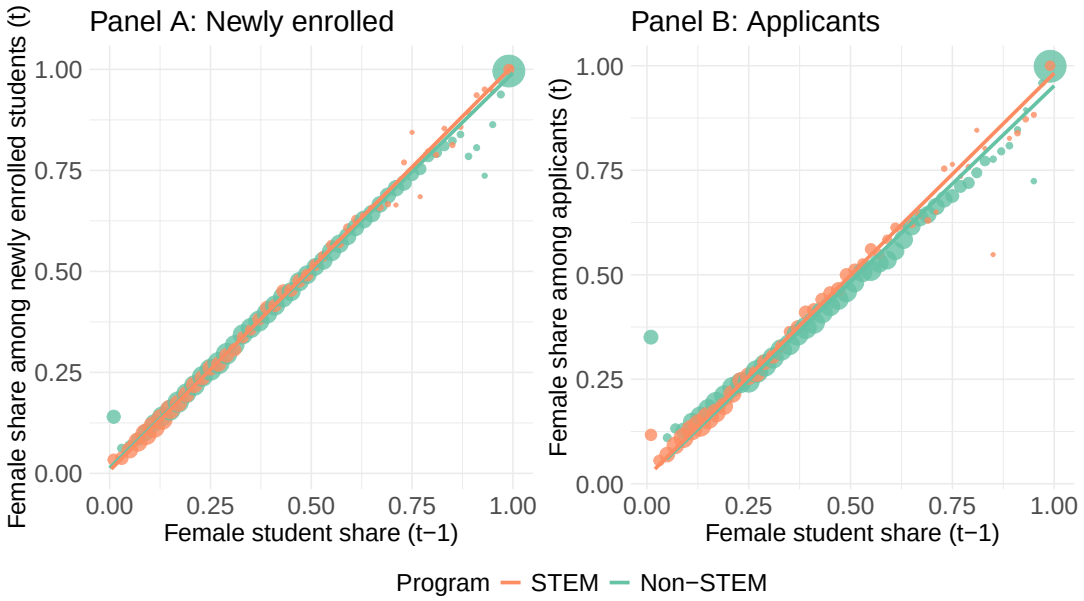
science (dark dotted), and reading (dark dash-dotted) score distributions among 15-year-old students in 2018 who planned to attend a four-year college, as well as the share of female students in STEM (orange) and non-STEM (green) college programs in 2023. The figure suggests potential inefficient allocation of talent, as in other OECD countries: although there are no significant gender differences in mathematics and science scores at age 15, significantly fewer female students choose STEM programs in college. While female students excel in reading, this advantage is not large enough to attribute the STEM gender gap to comparative advantage.[16]

Despite this potential inefficiency in talent allocation, gender ratios across college programs are surprisingly persistent. Figure 2 plots binned averages of the female share among newly enrolled students (Panel A) and among applicants (Panel B) against the total female student share in the same program in the previous year. Panel A uses the universe of STEM (orange, N=594) and non-STEM (green, N=1,409) college programs in Japan observed over 21 years (2003–2023). Panel B uses applicant data available from 2015 onward (9 years, N=507 for STEM, N=1,212 for non-STEM).

The figure shows that programs attract roughly the same share of female applicants and new

16. While the differences between female and male students are less pronounced in Japan than in the OECD average, the general patterns are the same; see Appendix Figure B1.

Figure 2: Persistence of the Gender Ratio Across College Programs



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students as the previous year's female share, making gender composition highly persistent. This pattern holds for both STEM and non-STEM programs. Two competing explanations exist for this persistence: taste heterogeneity and state dependence (see, for example, Heckman 1981; Hyslop 1999). Taste heterogeneity implies that female students are inherently less inclined to pursue STEM (and male students less inclined to pursue non-STEM), resulting in persistent gender ratios regardless of current composition. State dependence implies that students avoid programs where they expect to be a minority, meaning current composition causally affects future composition. While observationally equivalent, these explanations have starkly different policy implications: under taste heterogeneity, interventions to shift the female share will not break persistence; under state dependence, they will. Our experiment is designed to distinguish between these two competing explanations.

3 Experimental Design

To investigate whether the gender ratios at colleges affect high school students' college choices, we conducted an incentivized discrete choice experiment at four selective academic high schools in the Greater Tokyo area in Japan. We integrated this experiment as a "career planning module" within the 10th-grade curriculum of the participating high schools from December 2023 to July 2024. The experiment was conducted in person at three high schools and asynchronously online at one high

school.

The experiment lasted about 40 minutes on average, including distributing the participation gifts. A total of 628 students took part, with 619 providing valid responses (311 female, 298 male, 10 non-binary). Since this study focuses on binary gender, we excluded responses from non-binary students, resulting in 609 responses with 15 choices each, for a total of 9135 choices.

3.1 Sample Selection

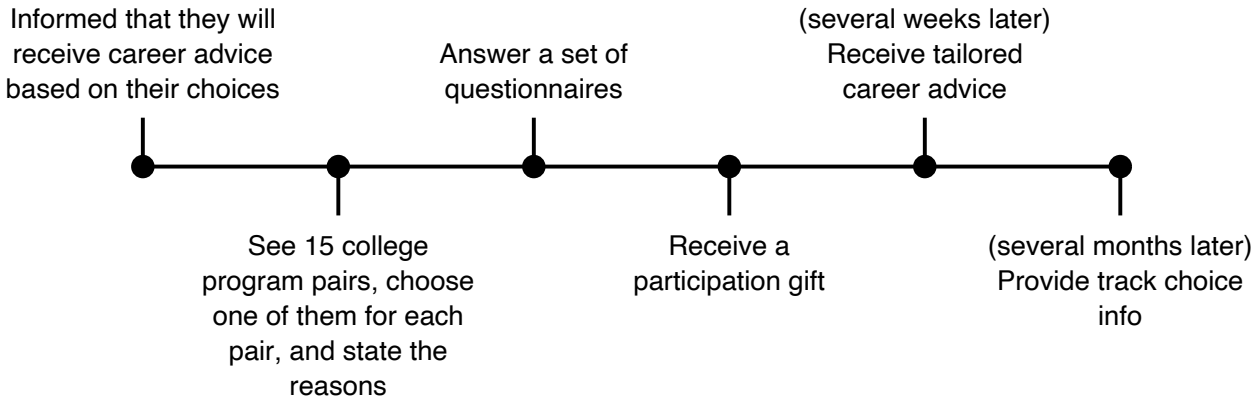
Schools We contacted teachers at academic high schools in the Greater Tokyo area through our network and obtained their consent to conduct the experiment as part of their school curriculum within a career guidance class. This was intended to make the experiment feel natural to students and to ensure dedicated time for participation. We restricted our potential sample to academic high schools in the Greater Tokyo area for three reasons. First, we wanted to ensure the experimental content was relevant to students: the experiment focused on college choices, and students needed to be planning to attend college. Based on the schools' placement records, over 95% of the recent graduates attended college. Second, we wanted to include students whose mathematics skills did not constrain their major choices: as discussed in Section 2, most STEM programs require good mathematics abilities, and students from academic high schools are the primary group prepared for STEM. Third, we wanted to prevent students from implicitly considering potential location and financial constraints when making their choices: as discussed in Section 2, the Greater Tokyo area offers the widest variety of college programs, making such constraints less relevant there. We also wanted the attributes of the hypothetical college programs to appear natural to students, and the Greater Tokyo area was suitable for this purpose as well.

Students The teachers at the participating high schools distributed the information letter and consent form to guardians of all 10th-grade students, except at one school where only one class participated. We restricted the sample to 10th-grade students before their track choice because the track choice restricted the college programs they could apply for, as discussed in Section 2. The information letter did not mention that the experiment was about gender ratios or STEM to minimize the experimenter demand effect (Zizzo 2010). Instead, we explained that students would evaluate 15 hypothetical college programs, answer a short questionnaire, and receive a tailored career advice sheet based on their evaluations. We also clarified that the data from the experiment would be used for academic research to improve education policy. Nearly 90% of the guardians and students provided consent and participated.

3.2 Flow of the Experiment

Students were first told that they would receive a career advice sheet based on their choices in the experiment. We promised them that their responses and the career advice sheet would not be shared with anyone, including guardians, teachers, or peers, to minimize their potential influence, as previous studies suggest that they can affect students' study choices (Carlana 2019; Carlana and

Figure 3: Flow of the Experiment



Notes: This figure shows the flow of the experiment.

Corno 2024; Giustinelli 2016; Miserocchi 2024; Müller 2024; Valdebenito 2023). The students then saw 15 program pairs one by one and chose the one they would prefer to attend; see Figure 4 for an example of a hypothetical program pair. Additionally, for each pair, they indicated a program that applied better to each of four statements. The statements were developed prior to the experiment, based on an open-ended questionnaire in a pre-test conducted with a group of students different from the participants in this experiment about why they care about the gender ratios.

After making choices for the 15 program pairs, students completed a questionnaire about their demographics, academic abilities, behavioral traits, and beliefs about the gender ratios in average college programs in Japan. Appendix Section C presents the questionnaire. Afterward, students received a participation gift (a set of decorative, functional pens) equivalent to 500 JPY (approx. 5.27 USD in 2022 PPP) for their participation.[17]

Several weeks later, each student received a career advice sheet we created. The sheet had two parts: a tailored part and a non-tailored part. The tailored part included the top three attributes that students cared about most, along with the top reason they prioritized when choosing a college program, based on their choices in the hypothetical programs. The non-tailored part contained non-individualized information useful for most high school students, such as tips for choosing college programs, college admissions, financing, studying abroad, attending graduate school, and finding jobs.

Finally, several months after the experiment, we requested students' track choice data from schools. Some schools provided data directly from their records; others provided it through a follow-up survey completed by students.[18] In total, we obtained valid responses from 461 out of 609 students (or 75.7% of students).

17. USD to JPY PPP was 94.93 in 2022: https://www.oecd.org/en/data/indicators/purchasing-power-parities-ppp.html (accessed November 18, 2024).

18. Approximately 8-15 months. It is 15 months because one school had their track choice at the end of the 11th grade.

Figure 4: Hypothetical Program Pair

Pair 4/15

AB College Dept. of Literature		AX College Dept. of Engineering
	Dept. Characteristics	
57.5	Selectivity index	62.5
700	Cohort size	600
35% male, 65% female	Student gender ratio	55% male, 45% female
	College Characteristics	
Yes	Intl exchange program	No
65%	Club participation rate	45%

Which program would you like to attend?

AB College, Dept. of Literature	AX College, Dept. of Engineering

Which program do you feel these statements apply to more?

	AB College, Dept. of Literature	AX College, Dept. of Engineering
I can do well in my studies	○	○
I can find a career I want to pursue	○	○
I can fit in	○	○
I can meet inspiring seniors	○	○

Notes: This figure shows an example of a hypothetical program pair students would see during the experiment.

3.3 Attributes

We randomly assigned attributes to each program, including college name, department, department selectivity index, department cohort size, department student gender ratio, whether the college has an international exchange program, and college club activity participation rate. These are attributes the students can find in college program guidebooks available at the school counselor's room, which they can read at any time.[19]

Our main attributes of interests are (i) student gender ratio and (ii) department, which indicates whether the program is STEM or non-STEM. We included other attributes to make the programs appear more realistic to students and selected attribute value ranges that are plausible for students in our sample to reduce hypothetical bias (List and Shogren 1998; List, Sinha, and Taylor 2006). We asked them to assume that attributes not shown were identical between the programs.

College names consist of two uppercase letters and we draw them without replacement for each program in a pair from a list ranging from AA to BD to make sure they were unrelated to the actual college names. The department was drawn from a list of 12 popular departments among college students, of which 6 were STEM and 6 were non-STEM.[20] First, we randomly assigned either STEM or non-STEM to one program in the pair. If STEM was selected, then the other program was assigned non-STEM with 75% probability and STEM with 25% probability to reduce the chances that both programs are in the same category. Next, we selected a specific department within the relevant list. STEM departments included Physics, Chemistry, Biology, Engineering, Information Technology, and Agriculture. Non-STEM departments included Literature, Law, Business, Economics, Sociology, and Foreign Language. We excluded Medicine and Education, as both are popular but lead to specialized careers such as doctors, nurses, pharmacists, and teachers, which differ from most student career paths, and the attributes used may not be very relevant for these programs.

Other attributes include selectivity index, which ranges from 55 to 72.5 with an increment of 2.5; cohort size, which varies from 200 to 900 with an increment of 50; student gender ratio, which spans from 5% to 95% for females and males but sums to 100%; whether the college has an international exchange program, which has an 80% chance of being "Yes" and a 20% chance of being "No"; and club participation rate, which ranges from 40% to 85% with an increment of 5%.[21] These attribute values were drawn with replacement for each program. Appendix Table B3 shows the possible values for each attribute.

We used a hypothetical choice experiment because it allowed us to elicit students' preferences

19. Some of these guidebooks provide club participation as qualitative information. Its range is obtained from the National Federation of University Cooperative Associations (2023).

20. We first referred to the Ministry of Education, Culture, Sports, Science and Technology (MEXT) department-classification list (https://www.mext.go.jp/component/b_menu/other/___icsFiles/afieldfile/2018/08/02/1407357_4.pdf, accessed June 30, 2023) to compile a comprehensive set of programs. We then consulted a college-choice guide for high school students (https://shingakunet.com/journal/column/20210415000002/, February 21, 2023 version) and selected 12 popular programs.

21. We set the probability that a program has an international exchange program to 80% because most colleges in Japan have one.

over attributes that were varied independently. This design was crucial for three reasons: most STEM programs are male-dominated, leaving little variation in gender ratios; the gender ratio correlates with other program attributes; and using actual college names would lead students to infer attributes not shown.

3.4 Incentives

One concern with hypothetical choice experiments is that students may lack incentives to state their true preferences without real consequences. Although Hainmueller, Hangartner, and Yamamoto (2015) show that choices in hypothetical vignettes and actual behaviors are highly correlated, we address this concern using the incentivized resume rating method (Kessler, Low, and Sullivan 2019), which involves providing career advice based on students' choices.[22] Assuming students believed that we researchers held potentially valuable new information, the method is incentive compatible: the expected value of the advice increases with the truthfulness of their choices. Because students were from academic high schools and planned to attend selective colleges, we assume most believed the researchers had valuable academic and career information.[23]

Specifically, we provided the following information in the information letter and at the beginning of the experimental instructions. This wording closely followed the original incentivized resume rating studies (Kessler, Low, and Sullivan 2019; Low 2024):

> *Through this module we will give you information relevant for your career choice. You will complete it on the internet using a laptop or a tablet. It is expected to last for 25 minutes and consists of two parts:*
>
> - *Evaluation of the hypothetical 15 program pairs*
> - *A short questionnaire*
>
> *We will send you a career advice sheet created based on your evaluation.*

4 Data

4.1 Variable Construction

Academic Abilities We convert students' academic abilities, obtained through a post-experimental questionnaire, into population z-scores to make them comparable across schools and interpretable within the entire student pool. To do this, we use the latest placement records of graduates from

22. Several studies employed the incentivized resume rating method to elicit preferences for attributes that are hard to obtain from revealed preferences. Low (2024) elicited heterosexual adults' preferences for dating partners by providing dating advice from a dating coach based on their ratings of hypothetical opposite-gender partner profiles. Macchi (2023) elicited loan officers' preferences for borrowers by offering referrals to loan clients based on their ratings of hypothetical borrower profiles. Gallen and Wasserman (2023) elicited college students' mentor preferences by providing mentor characteristics that students care most about. Chan (2024) elicited patients' preferences for doctors by offering booking options based on patients' choices.

23. At the time of the experiment, Inoue obtained a PhD from one of Japan's most prestigious colleges, Saito earned a master's degree in the US and has rich industry experience, and Takahashi earned a PhD from a European university.

each participating high school, assign the selectivity index to each college in the records, and rank the placements by the selectivity index. We then assign each student a selectivity index based on their academic rank within their high school in specific subjects (reading, mathematics, English, and total). The selectivity index for each program is obtained from the list prepared by Kawaijuku in 2024, one of the most popular commercial college exam preparation companies in Japan.[24] Since public colleges require a larger number of subjects in the entrance exam, we add 0.5 to the index of public colleges (equivalent to 5 points in the raw index), following Araki, Kawaguchi, and Onozuka (2016).

Behavioral Traits We elicit students' behavioral traits through a post-experimental questionnaire: confidence in reading, mathematics, and English; competitiveness; and risk-taking, all rated on a 5-point Likert scale with 3 being neutral.[25] For confidence questions, we ask how accurately their recent exam scores, entered on the previous page, reflected their ability. We convert these 5-point scales to the range [-1, 1] for better interpretability, with 0 being neutral.

4.2 Summary Statistics

Table 1 presents summary statistics for 311 female and 298 male students in the final sample (totaling 609 students), along with their differences. Panel A presents students' demographics and indicates no differences in parents' education levels or parental investments (proxied by extra schooling days per week) between female and male students. Panel B presents students' academic abilities and shows that female students outperform male students in reading and English, while male students excel in mathematics; female students have a slight edge in overall scores.[26] As expected given our sample of academic high schools, students perform approximately 0.6 to 0.7 standard deviations above the national average. Panel C presents students' behavioral traits, showing that male students are more confident in their mathematics and English abilities – despite lower English performance – and are less risk-averse than female students, consistent with existing literature on gender differences in preferences (Croson and Gneezy 2009). Although statistically insignificant, the gender difference in competitiveness is also consistent with existing literature, with male students more competitive than female students.

5 Gender Ratio Affects Students' College Choices

5.1 Econometric Framework

Estimation of Preferences To estimate students' preferences for program attributes, we assume that student i of gender g's preferences over program d with attributes X in pair j are represented

24. https://www.keinet.ne.jp/exam/ranking/index.html (accessed December 18, 2024)
25. The questionnaire-based competitiveness measure is adapted from Buser, Niederle, and Oosterbeek (2024), and the risk-taking measure is adapted from Dohmen et al. (2011).
26. Appendix Figure B2 shows the distribution of abilities for female and male students.

Table 1: Summary Statistics of Students in the Final Sample

	Female (N=311)	Male (N=298)	Difference (M – F)
Panel A: Demographics			
Mother bachelor or above	0.59	0.56	-0.03
	(0.49)	(0.50)	(0.04)
Father bachelor or above	0.80	0.77	-0.03
	(0.40)	(0.42)	(0.04)
Both bachelor or above	0.50	0.48	-0.02
	(0.50)	(0.50)	(0.04)
Extra schooling (no. days/week)	0.94	0.92	-0.02
	(1.06)	(1.04)	(0.09)
Panel B: Academic abilities (population z-score)			
Reading score	0.82	0.59	-0.23***
	(0.74)	(0.75)	(0.06)
Math score	0.65	0.77	0.12**
	(0.75)	(0.75)	(0.06)
English score	0.82	0.58	-0.25***
	(0.69)	(0.78)	(0.06)
Total score	0.71	0.61	-0.11*
	(0.74)	(0.76)	(0.06)
Panel C: Behavioral traits			
Reading confidence [-1,1]	-0.01	0.05	0.06
	(0.39)	(0.49)	(0.04)
Math confidence [-1,1]	-0.02	0.18	0.20***
	(0.39)	(0.50)	(0.04)
English confidence [-1,1]	-0.00	0.10	0.10***
	(0.40)	(0.45)	(0.03)
Competitiveness [-1,1]	0.03	0.11	0.08
	(0.67)	(0.68)	(0.05)
Risk-taking [-1,1]	-0.36	-0.18	0.18***
	(0.61)	(0.70)	(0.05)

Notes: This table presents summary statistics of 311 female and 298 male students in the final sample (609 students in total) as well as their differences. Standard deviations and standard errors are in parentheses. Significance levels: * 10%, ** 5%, and *** 1%.

by a linear indirect utility function:

$$V_{ijd} = X'_{jd}\beta^g + \varepsilon_{ijd} \tag{1}$$

The probability that student i chooses the right program r over left l in choice pair j is then:

$$P(V_{ijr} > V_{ijl}|X, g) = F((X_{jr} - X_{jl})'\beta^g) \tag{2}$$

where F is the cumulative distribution function (CDF) of $\varepsilon_{ijr} - \varepsilon_{ijl}$. We assume an identity function for the CDF, $F(x) = I(x) = x$, and estimate the model using OLS for ease of interpretation and decomposition. The empirical model we estimate is therefore:

$$C_{ij}^r = \alpha^g + \zeta^g(FShare_{jr} - FShare_{jl}) + (W_{jr} - W_{jl})'\omega^g + \epsilon_{ij} \tag{3}$$

where C_{ij}^r is an indicator variable equal to 1 if student i chooses the right program in choice pair j, $FShare_{jd}$ is the share of female students in program d within pair j, $W_{jd} \equiv X_{jd} \setminus \{FShare_{jd}\}$ is a vector of program d's attributes in pair j excluding the female student share, and α^g is the intercept for the right program.

We also present the results with logit (assume $F(x) = \Lambda(x)$) as a robustness check.

Decomposition of the Choices To investigate the underlying reasons for students' program choices, we treat the four reasons we elicited in the experiment as mediators: fit in, role model, studies, and career. We then decompose the treatment effects of the female student share on the choices into these four reasons, following Gelbach (2016) and Gong, Lu, and Song (2021).

Replace C_{ij}^r in equation 3 with the four reasons:

$$M_{ij}^m = \kappa^{m,g} + \xi_{m,g}(FShare_{jr} - FShare_{jl}) + (W_{jr} - W_{jl})'\psi^{m,g} + \nu_{ij}^m \tag{4}$$

where M_{ij}^m (for $m = 1, 2, 3, 4$) is an indicator variable equal to 1 if student i indicated that reason m better applies to the right program in the choice pair j.

Finally, include all the M_{ij}^ms in equation 3:

$$C_{ij}^r = \alpha^{g,full} + \zeta^{g,full}(FShare_{jr} - FShare_{jl}) + (W_{jr} - W_{jl})'\omega^{g,full} + \sum_{m=1}^{4} \eta^{m,g} M_{ij}^m + \epsilon_{ij}^{full} \tag{5}$$

Gelbach (2016) shows that:

$$\hat{\zeta}^g = \hat{\zeta}^{g,full} + \sum_{m=1}^{4} \hat{\eta}^{m,g}\hat{\xi}^{m,g} \quad \forall g \tag{6}$$

where $\hat{\eta}^{m,g}\hat{\xi}^{m,g}$ is the part of the treatment effects $\hat{\zeta}^g$ explained by reason M_{ij}^m, and $\hat{\zeta}^{g,full}$ is the part of the treatment effects $\hat{\zeta}^g$ unexplained by any of the four reasons.

We discretize the female student share as discussed later, so we perform this decomposition for each bin of the share. We first present the results of the preference estimations (Section 5.2), then apply this decomposition to understand underlying reasons (Section 5.3).

Table 2: Preferences for Program Attributes

Sample:	Female		Male		All	
Outcome:	Choose the program (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
STEM	-0.079***	-0.079***	0.038**	0.038**	0.038**	0.038**
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)
Female student share	0.090***	0.821***	-0.036	0.818***	-0.036	0.818***
	(0.022)	(0.088)	(0.024)	(0.089)	(0.024)	(0.089)
Female student share squared		-0.745***		-0.875***		-0.875***
		(0.088)		(0.090)		(0.090)
Selectivity index (population SD)	0.046***	0.047***	0.075***	0.075***	0.075***	0.075***
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Cohort size/100	0.003	0.003	0.006**	0.006**	0.006**	0.006**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Intl exchange program	0.059***	0.060***	0.033**	0.034**	0.033**	0.034**
	(0.014)	(0.014)	(0.013)	(0.013)	(0.013)	(0.013)
Club participation rate	0.193***	0.196***	0.122***	0.130***	0.122***	0.130***
	(0.042)	(0.042)	(0.041)	(0.040)	(0.041)	(0.040)
Female					0.016	0.015
					(0.011)	(0.011)
STEM x Female					-0.117***	-0.116***
					(0.024)	(0.024)
Female student share x Female					0.126***	0.003
					(0.032)	(0.126)
Female student share squared x Female						0.131
						(0.126)
Selectivity index (population SD) x Female					-0.029*	-0.028*
					(0.016)	(0.016)
Cohort size/100 x Female					-0.003	-0.003
					(0.004)	(0.004)
Intl exchange program x Female					0.027	0.026
					(0.019)	(0.019)
Club participation rate x Female					0.071	0.066
					(0.058)	(0.058)
Constant	0.503***	0.503***	0.488***	0.487***	0.488***	0.487***
	(0.007)	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)
Adj. R-squared	0.036	0.054	0.021	0.045	0.029	0.050
No. observations	4649	4649	4451	4451	9100	9100
No. students	310	310	297	297	607	607

Notes: This table presents the OLS coefficient estimates on the program attributes with choice as the dependent variable. Columns 1 and 2 present estimates for female students, columns 3 and 4 present estimates for male students, and columns 5 and 6 present estimates for differences between female and male students. Standard errors are clustered at the student level. Significance levels: * 10%, ** 5%, and *** 1%.

5.2 Preferences for the Gender Ratio

Table 2 presents preferences for program attributes.[27] First, female students are 7.9 percentage points less likely to choose STEM programs, while male students are 3.8 percentage points more likely to do so, consistent with the literature. Second, both female and male students prefer

27. Appendix Table B4 presents the same specifications but with indicator variables for the four reasons instead of choice as the outcome variables.

programs with higher selectivity indices, but male students show a slightly stronger preference: a one-standard-deviation increase in selectivity raises female students' choice probability by 4.7 percentage points and male students' by 7.5 percentage points – a 2.8 percentage points gender gap.

Third, students also favor the social aspects of the programs, consistent with the evidence that students value non-academic amenities in college (Ersoy and Speer 2025; Gong et al. 2021; Jacob, McCall, and Stange 2018). Both female and male students prefer programs that have an international exchange program: having an international exchange program increases female students' choice probability by 6.0 percentage points and male students' choice probability by 3.4 percentage points. Club participation rates also affect students' choices: a 10 percentage points increase in club participation rate increases female students' choice probability by 2.0 percentage points and male students' choice probability by 1.3 percentage points. The effect of the cohort size is quantitatively minimal: an increase of a program size by 100 students increases male students' choice probability by 0.6 percentage points but has no statistically significant effect on female students. Logit estimates in Appendix Table B5, where we convert the coefficient estimates into average marginal effects show essentially the same results as those with OLS.

The effect of female student share is highly non-linear: the coefficient on the squared term is significant for both genders (columns 2 and 4). We therefore discretize the female share into 19 bins and re-estimate with 50% as the baseline.

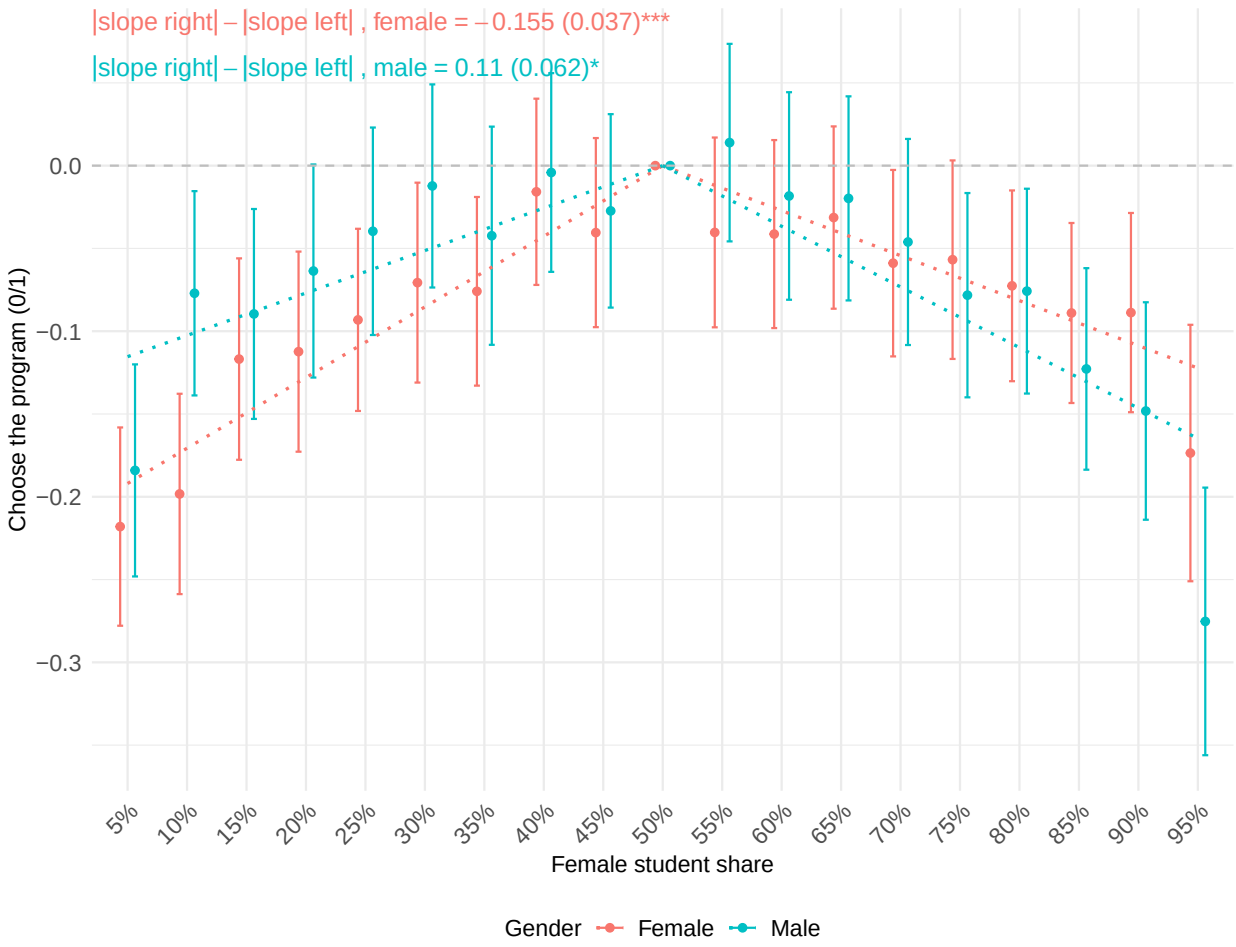
Figure 5 reveals three patterns. First, both genders prefer gender-balanced programs over those where they would be a minority: relative to 50% female, being a minority (25% own gender) reduces choice probability by 11 percentage points for female students and 9 percentage points for male students. Second, both genders also prefer balance over being a strong majority: being 75% own gender reduces choice probability by 6.5 percentage points for both. Third, despite this preference for balance, students still prefer majority to minority status – the fitted slopes are steeper on the minority side than the majority side, indicating stronger aversion to minority status than to majority status.

5.3 Underlying Reasons

Figure 6 decomposes the effect of gender composition into four stated reasons – fitting in, finding a role model, doing well in studies, and finding a career – plus an unexplained residual.

Panel A shows that the main reason the gender ratio affects female students' program choices varies significantly depending on whether most students in the program are male or female. When the majority of students are male, the gender ratio affects female students' program choices primarily due to their concerns about fitting in. On the other hand, when the majority of students are female, multiple factors – not only concerns about fitting in, but also concerns about finding a role model, doing well in studies, finding a career, and unexplained reasons – all contribute to how the gender ratio affects their choices. One notable observation is that the unexplained part is larger when the majority of students are female. It may reflect marriage market concerns, as students often find their future partners in the field of study within their college (Artmann et al. 2021; Kirkebøen

Figure 5: Preferences for the Gender Ratio



Notes: This figure plots the coefficient estimates and the 95% confidence intervals of female student share discretized into 19 equally spaced bins, with 50% as the baseline, separately for female students (red) and male students (blue). Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for female and male students on both sides (below 50% and above 50%). We imposed the constraint that the lines pass through the 50% point.

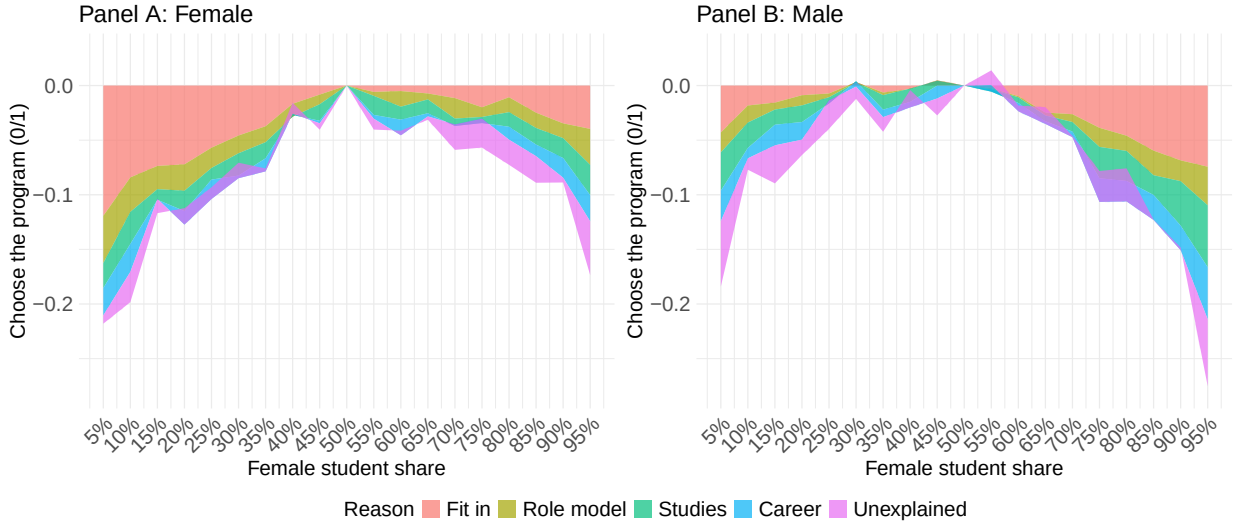
et al. 2025; Pestel 2021), especially in fields with a larger fraction of opposite-gender peers (Artmann et al. 2018).

Interestingly, Panel B shows that the patterns are very similar for male students: when most students are female, the gender ratio affects male students' program choices primarily through the expected difficulty in fitting in. When most students are male, however, all four reasons, along with other reasons beyond these four, explain their choices.

5.4 Heterogeneity of Preferences

Figure 7 plots coefficient estimates and the 95% confidence intervals for female and male students, where we interact the attributes with the STEM dummy. Panels A and B show all students, Panels C and D restrict the sample to students among the top 50% in mathematics abilities in the sample,

Figure 6: Decomposition of Preferences for the Gender Ratio



Notes: This figure plots $\hat{\eta}^{m,g}\hat{\xi}^{m,g}$ $(m = 1, 2, 3, 4)$ and $\hat{\zeta}^{g,full}$ from equation 6 estimated for each of the 19 bins separately for female (Panel A) and male (Panel B) students. The red area shows fitting in, the yellow area shows finding a role model, the green area shows doing well in studies, the blue area shows finding a career to pursue, and the purple area shows reasons other than these four.

and Panels E and F restrict the sample to students among the top 50% in reading abilities in the sample.[28]

Panel A shows that female students' preferences for gender composition are similar in STEM and non-STEM programs. Panel B reveals a different pattern for male students: they are less sensitive to gender composition in STEM than in non-STEM. Being a minority reduces male students' choice probability by 8 percentage points in STEM versus 12 percentage points in non-STEM; being a strong majority reduces it by 6 percentage points in STEM versus 8 percentage points in non-STEM. Panels C and D restrict the sample to students in the top half of mathematics ability. Here, female students exhibit the opposite pattern from males: they show stronger minority avoidance in STEM than non-STEM. Relative to a gender-balanced program, being a minority (25% female) reduces high-math-ability female students' choice probability by 14 percentage points in STEM versus 7 percentage points in non-STEM – a twofold difference. High-math-ability male students show no significant STEM/non-STEM heterogeneity (Panel D).[29]

Panels E and F examine students in the top half of reading ability; neither gender shows significant heterogeneity by field. Preferences for other attributes also do not differ between STEM and non-STEM (see Appendix Figure B7).[30]

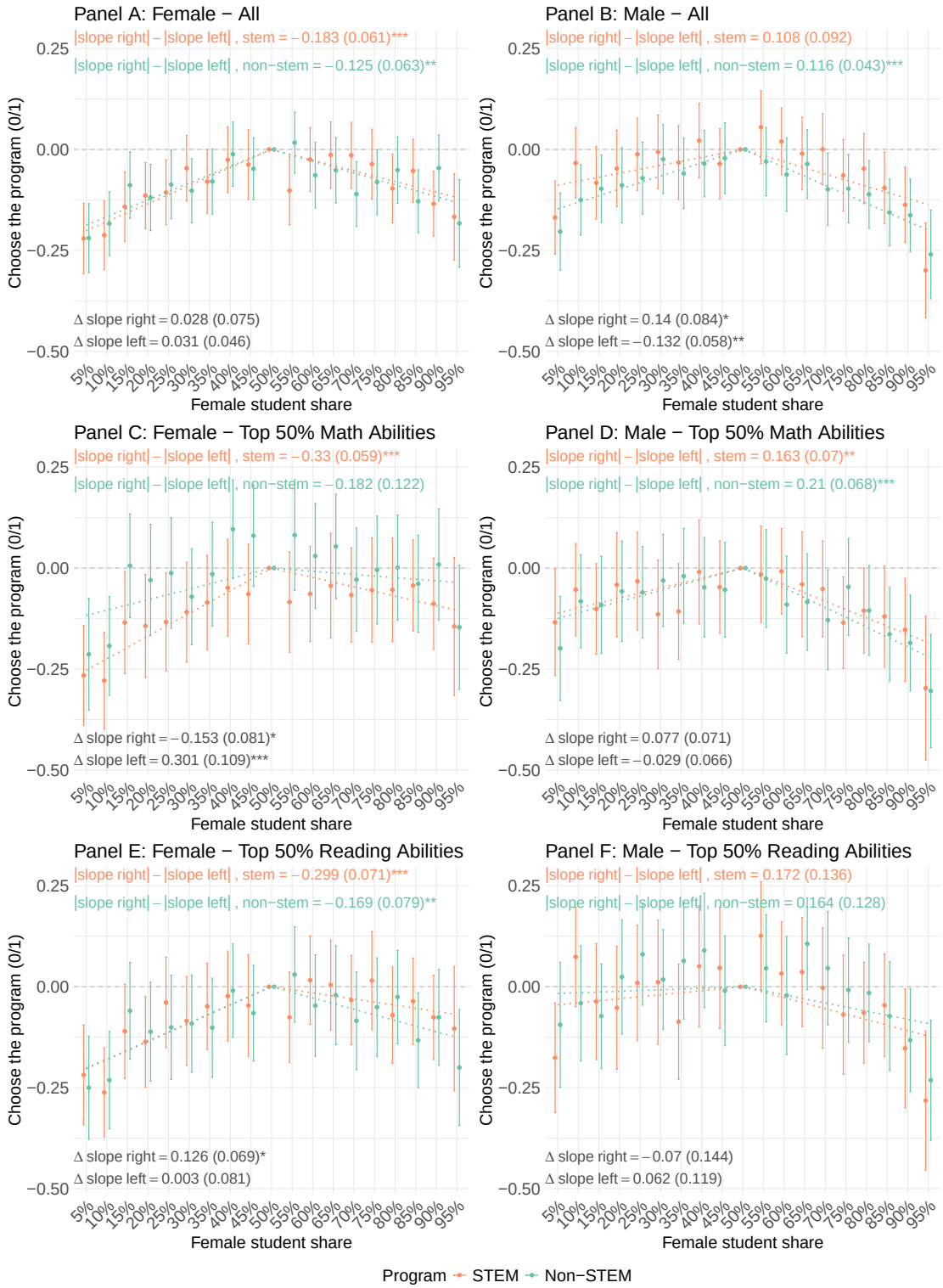
In summary, both genders prefer balance to majority and majority to minority. But sensitivity

28. There is no significant heterogeneity by other program attributes; see Appendix Figures B3 and B4.

29. Appendix Figure B5 presents differences in slopes between students in the top and bottom 50% of mathematics and reading abilities. These students also differ in their preferences for other attributes, as shown in Appendix Figure B6.

30. These ability differences are not coming from differences in parental education level, as shown in Appendix Figure B8.

Figure 7: Heterogeneity of Preferences for the Gender Ratio – STEM vs. Non-STEM



Notes: This figure plots coefficient estimates and the 95% confidence intervals for female and male students, where we interact the attributes with the STEM dummy. Panels A and B plot all female and male students' preferences, Panels C and D plot female and male students in the top 50% of mathematics ability in the sample, and Panels E and F plot female and male students in the top 50% of reading ability in the sample. Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for each side (below 50% and above 50%) for two groups. We imposed the constraint that the lines pass through the 50% point. The differences in the slopes between STEM and non-STEM are calculated using the weighted least squares standard errors. Significance levels: * 10%, ** 5%, and *** 1%.

to gender composition varies by gender, ability, and field: high-mathematics-ability female students are more deterred by minority status in STEM than in non-STEM, while male students are less sensitive to composition in STEM overall. One interpretation is that students perceive STEM as a male domain where male students feel they belong regardless of the gender ratio, whereas female students – especially those accustomed to male-dominated academic environments – recognize that fitting in requires a critical mass of same-gender peers.

6 Quantification of Inefficiency in Talent Allocation

Section 5.4 documents that preferences vary by gender, ability, and field. High-mathematics-ability female students exhibit stronger minority avoidance in STEM than in non-STEM, while male students show the opposite pattern – weaker sensitivity to gender composition in STEM overall. These preferences, combined with the low female share in STEM, imply potential inefficiency in talent allocation. This section quantifies that inefficiency.

We compare two scenarios in terms of how talent allocation changes between them:

- The optimal scenario: students have no preferences for the gender ratio. We use this as a benchmark for the best attainable allocation.[31]
- The baseline scenario: students have preferences for the gender ratio as estimated from the data, and the female share is 25% in STEM and 50% in non-STEM. We use this as a realistic baseline.

This comparison requires estimates of each student's preferences for program attributes, especially for the female share and STEM. We estimate these using a mixed logit model.

6.1 Econometric Framework

Estimation of Individual-Level Preferences Rewrite equation 2 as follows:

$$P_{ijd}^{g} = \frac{\exp(X_{jd}'\beta^{g})}{\sum_{k\in\{r,l\}}\exp(X_{jk}'\beta^{g})} \tag{7}$$

where P_{ijd}^{g} is student i of gender g's choice probability of program $d \in \{r, l\}$ with attributes X in pair j. This expression is equivalent to equation 2 in Section 5 (see, for example, Train 2009, Section 3.1). Unlike equation 2, however, we allow interactions between STEM and the female share to capture the heterogeneity documented in Section 5.4. We use a quadratic functional form for the female share to keep the number of parameters adequate for the sample size and to define choice probabilities over the continuous female share.

Now, let $\beta_i^g = (\alpha_i^g, \gamma^g)'$ and assume α_i^g is a random variable with density $f(\alpha^g|\theta^g)$, where θ^g are parameters of this distribution for gender g. We keep γ^g as fixed, as before. Partition $X_{jd} = (X_{jd}^1, X_{jd}^2)'$, where $X_{jd}^1 = (STEM_{jd}, FShare_{jd}, FShare_{jd}^2, STEM_{jd} \times FShare_{jd}, STEM_{jd} \times$

31. Setting the STEM female share to 50% instead of removing gender preferences yields essentially the same results.

$FShare^2_{jd})'$. Then the choice probability can be written in a mixed logit form:[32]

$$P^g_{ijd}|\theta^g = \int \frac{\exp((X^{1'}_{jd}, X^{2'}_{jd})(\alpha^g_i, \gamma^g)')}{\sum_{k\in\{r,l\}} \exp((X^{1'}_{jk}, X^{2'}_{jk})(\alpha^g_i, \gamma^g)')} f(\alpha^g|\theta^g) d\alpha^g \tag{8}$$

where we allow arbitrary correlations among elements in α^g_i except the parameter for STEM. We assume that the mixing distribution f follows a normal distribution, as is standard (Train 2009).[33] We approximate the integral using 4,000 Halton draws.[34] Continuous variables are normalized for model stability.

After estimating the mixed logit model, we obtain individual-specific parameters α^g_i as posterior means from their conditional distribution, following Train (2009):

$$h(\alpha^g|y_i, X^1_i, g, \theta^g) \tag{9}$$

where y_i is a vector of i's choices across the 15 pairs, and X^1_i is a matrix of attributes in X^1_{jd} that i observed across the 15 pairs.[35]

6.2 Utility Cost

We first examine the utility cost of a low female share in STEM among students who prefer STEM. Panel A of Appendix Figure B9 shows that, compared to the optimal scenario, female students incur a utility cost of 0.39 and male students 0.15 under the baseline scenario. As shown in Panel B, this is equivalent to 31.6% of the STEM preference for female students and 9.8% for male students.[36] This is a substantial cost: 15.2% of female students and 4.0% of male students forgo STEM due to the low female share.

6.3 Inefficiency in Talent Allocation

We now examine how the low female share in STEM affects talent allocation.

Since both female and male students incur utility costs from a low female share in STEM, removing gender-ratio preferences in the optimal scenario allows more students to enter STEM. However, when enrollment differs across scenarios, comparing ability sorting becomes difficult. To

32. Mixed logit relaxes the following standard logit assumptions: (i) no random preference variation among individuals, (ii) independence of irrelevant alternatives, and (iii) no correlation in unobserved factors over time (Train 2009, Section 6). Our logit results in Table 2 remain valid as there are only two alternatives in each pair and no notable time-varying unobserved factors in the experiment. We use mixed logit to relax the first assumption and estimate individual-level preference parameters.

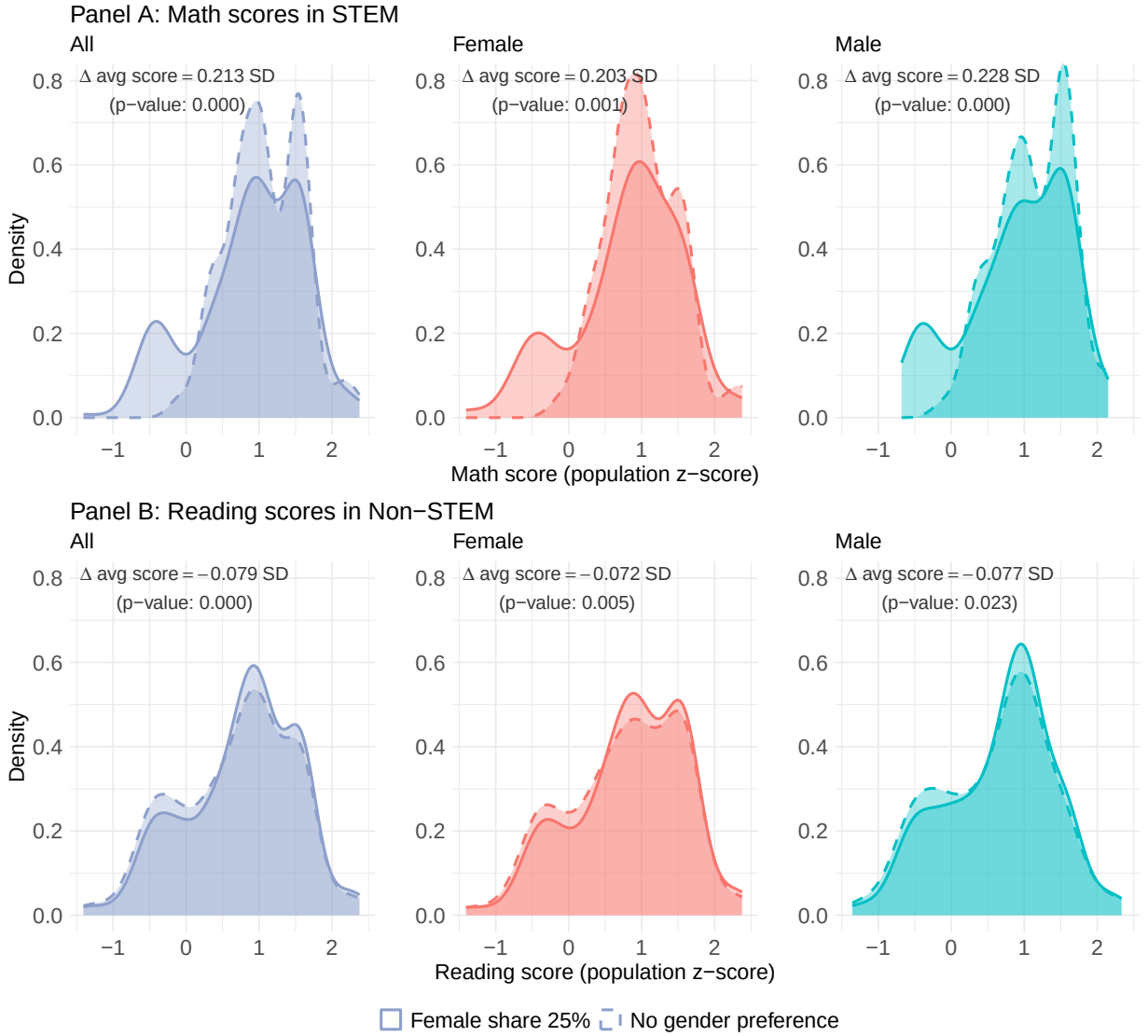
33. Appendix Section A discusses the mixed logit model performance and variation in individual-level preference parameters.

34. We use Halton draws instead of pure Monte Carlo draws for stability and faster convergence (Train 2009, Section 9).

35. Individual-specific parameters from mixed logit are used in other papers. For example, León and Miguel (2017) estimate the value of a statistical life among Africans and non-Africans using individual-level parameters. Kremer et al. (2011) estimate the value of spring water protection in Kenyan households using individual-level parameters.

36. Panel B excludes the bottom 10% of students whose STEM preference is weak, as they inflate the willingness to pay.

Figure 8: Changes in the Ability Distributions



Notes: This figure plots students' ability distributions under the baseline scenario (solid) and the optimal scenario (dashed). Panel A shows the distributions for STEM and Panel B for non-STEM. Gray plots show distributions for all students, red for female students, and green for male students. The change in average scores from the baseline to the optimal scenario, along with bootstrapped p-values, is presented in the top left of each plot.

make a meaningful comparison, we impose capacity constraints in the optimal scenario, setting each field's capacity equal to its enrollment under the baseline scenario. We allocate students using the student-proposing deferred acceptance algorithm (Gale and Shapley 1962).[37]

The algorithm works as follows:

1. Students apply to their most preferred field (STEM or non-STEM) according to their prefer-

37. The student-proposing deferred acceptance algorithm is strategy-proof and stable. Although alternative algorithms such as the Boston mechanism and Top Trading Cycles often yield higher welfare, the gains typically arise from institutional or cognitive constraints (see, e.g., Calsamiglia, Fu, and Güell 2020; Calsamiglia, Haeringer, and Klijn 2010).

ences.

2. Each field ranks students by subject-specific ability – mathematics scores for STEM, reading scores for non-STEM – and tentatively admits the top-ranked students up to capacity, rejecting the rest.
3. Rejected students apply to their next preferred field.

Steps 2 and 3 repeat until no further rejections occur. Capacity for each field is set equal to the number of students allocated to that field under the baseline scenario without capacity constraints (i.e., when all students are admitted to their most preferred field).

Figure 8 plots students' ability distributions under the baseline scenario (solid) and the optimal scenario (dashed). Panel A shows the distributions for STEM and Panel B for non-STEM. Gray plots show distributions for all students, red for female students, and green for male students. The change in average scores from the baseline to the optimal scenario, along with bootstrapped p-values, is presented in the top left of each plot.

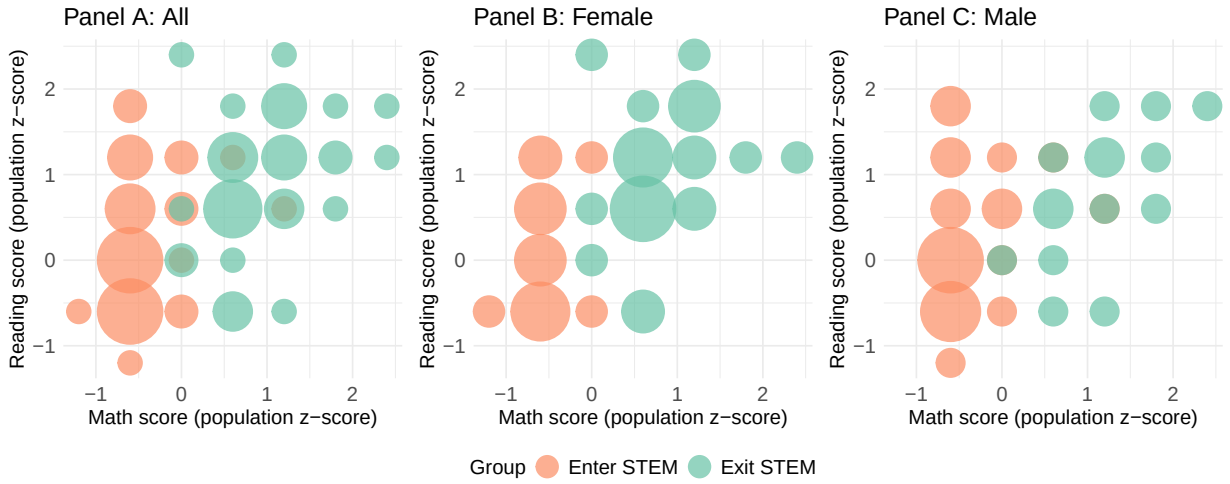
Panel A shows that the average STEM student under the baseline scenario has a mathematics score 0.2 standard deviations lower than under the optimal scenario; the difference is highly statistically significant. This lower score reflects inefficient talent allocation: some female students with mathematics scores 1 standard deviation above the population mean, and some male students with scores 1–2 standard deviations above the mean, apply to non-STEM under the baseline scenario. Their seats are instead allocated to female and male students whose mathematics scores are 0.5–1 standard deviations below the mean.

Panel B shows that, for non-STEM, the average student under the baseline scenario has a reading score 0.08 standard deviations higher than under the optimal scenario; this difference is also highly statistically significant. Thus, allocation in non-STEM is less efficient under the optimal scenario – unlike in STEM. However, the efficiency loss in non-STEM is only about one-third the size of the efficiency gain in STEM, so overall allocation improves.

To better understand which students switch from STEM to non-STEM – and from non-STEM to STEM – due to the low female share in STEM, Figure 9 plots students who exit (green dots) and enter (orange dots) STEM under the baseline scenario. Dot size indicates the number of students in each cell.

Panel A shows that, overall, students who exit STEM lie roughly on the 45-degree line – that is, they excel in both mathematics and reading. Students who enter STEM, however, do not excel in mathematics. Panels B and C show similar patterns for female and male students, though more female students exit STEM than male students – consistent with the stronger minority avoidance in STEM among high-mathematics-ability female students documented in Section 5.4. Thus, the inefficiency in talent allocation observed in Figure 8 arises because some students who excel in both subjects – and hence have more freedom to choose – move to non-STEM, while the capacity they vacate is filled by students who do not excel in mathematics.

Figure 9: STEM Entries and Exits Under the Baseline Scenario



Notes: This figure plots students who exit (green dots) and enter (orange dots) STEM under the baseline scenario. Dot size indicates the number of students in each cell.

6.4 Allocations Across Programs

To examine how allocations change across actual college programs, we compare student allocations under actual female shares with those under the optimal scenario. We focus on STEM and non-STEM programs in the Greater Tokyo area at colleges above a selectivity threshold that students in our sample can realistically pursue.[38]

We use the same student-proposing deferred acceptance algorithm as in Figure 8, with three modifications. First, we calculate the total number of seats in each program as the total number of students in our sample multiplied by that program's cohort size relative to the sum of all programs' cohort sizes. We then convert the resulting seats to integers using the largest remainder method so that capacities sum exactly to the sample size. Second, because capacity is determined this way, we do not admit everyone to their most preferred program even under actual female shares. Third, due to the small capacity in each program, we draw 2,000 independent shocks from a Type I extreme value distribution for each student's utility, compute 2,000 allocations, and use the average as each student's allocation.

Appendix Figure B10 plots student allocations across programs under actual female shares. Orange dots are STEM programs and green dots are non-STEM programs. As expected, the female share is consistently above 50% in non-STEM programs and consistently below 50% in STEM programs. Moreover, the more selective the program, the lower the female share, especially among STEM programs – likely reflecting high-mathematics-ability female students' greater minority avoidance in STEM as well as male students' stronger preferences for selective programs.

38. We use data from the School Basic Survey for program attribute values. Club participation rate is set to the median value in the experiment. We exclude Sophia University's Theology program from non-STEM as it is highly specialized.

Figure 10: Changes in Allocation Across Programs

Notes: This figure plots changes in student allocations across programs in the Greater Tokyo area from actual female shares to the optimal scenario; each dot represents a program. Orange dots are STEM programs and green dots are non-STEM programs. We use the same student-proposing deferred acceptance algorithm as in Figure 8 with a few minor alternations to take account for these programs' actual capacity. Due to the small capacity in each program, we draw 2,000 independent shocks from a Type I extreme value distribution for each student's utility, compute 2,000 allocations, and use the average as each student's allocation. Changes in average scores from the baseline to the optimal scenario, along with bootstrapped p-values (2,000 draws), are presented in the top left of each plot.

Figure 10 plots changes in allocations across programs from actual female shares to the optimal scenario, where students have no preferences over gender composition. The figure shows that, under the optimal scenario, more female students are allocated to STEM programs and more male students to non-STEM programs. The increase in female students is more pronounced in STEM programs with higher selectivity, consistent with high-mathematics-ability female students' greater minority avoidance in STEM. The decrease in male students is concentrated among less selective non-STEM programs, consistent with male students' weaker minority avoidance in non-STEM.

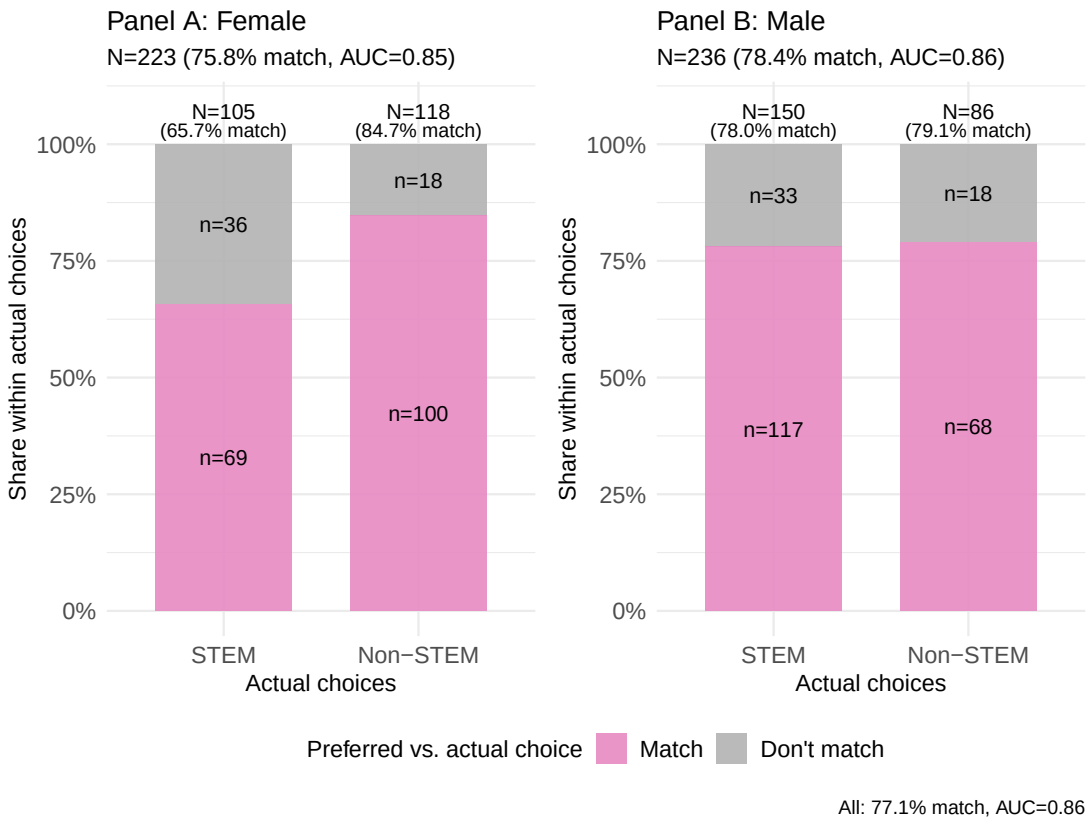
A caveat is that, unlike in Figure 8, this reallocation neither reduces nor improves efficiency: the average STEM student's mathematics score increases by 0.07 standard deviations, while the average non-STEM student's reading score decreases by roughly the same amount. This is likely because correlations among program attributes partially mitigate misallocation.

In summary, the low female share in STEM leads to inefficient talent allocation: students who excel in both mathematics and reading shift from STEM to non-STEM, with female students more likely to exit, while students who do not excel in mathematics fill their places. The resulting efficiency loss in STEM is only partially offset by a modest efficiency gain in non-STEM. Allocations

across actual programs are consistent with these results, though the overall inefficiency is smaller.

7 Preferences and Actual Choices

Figure 11: Preferences and Actual Choices



Notes: This figure shows the fraction of students who chose their preferred track. We classify a student's preferred track as STEM if the mixed logit coefficient estimate for the STEM dummy is positive, and non-STEM otherwise. We denote the sciences track as STEM and the humanities track as non-STEM.

Finally, we examine whether students choose their preferred track. Figure 11 shows the results. We classify a student's preferred track as STEM if the mixed logit coefficient for the STEM dummy is positive, and non-STEM otherwise; we denote the sciences track as STEM and the humanities track as non-STEM.

Overall, 77.1% of students chose their preferred track (75.8% for female students, 78.4% for male students). We assess classification quality using the ROC-AUC, which measures the model's ability to discriminate between STEM and non-STEM choosers (Hanley and McNeil 1982). The AUC indicates excellent discriminatory power (Hosmer and Lemeshow 2000, Chapter 5): 0.86 for all students, 0.86 for male students, and 0.85 for female students. This is notable given that AUC analysis typically evaluates contemporaneous signals and outcomes, whereas our data span several months between the experiment and actual track choices.

Second, female students who prefer STEM are 12.3 percentage points less likely to follow through than male students who prefer STEM. This gap is broadly consistent in magnitude with the estimated fraction of students who forgo STEM due to the low female share (Panel B of Appendix Figure B9). Importantly, this gap does not reflect differential prediction accuracy: we classify preferences using only the coefficient on the STEM dummy, which captures intrinsic field preferences but not preferences over gender composition. Since female share preferences are excluded from the classification, female students classified as "preferring STEM" may still avoid it due to the low female share – consistent with our main finding that female students anticipate difficulty fitting into male-dominated environments.

8 Discussion and Conclusion

Female students are significantly less likely to pursue STEM fields at colleges in OECD countries, despite the negligible gender gap in mathematics and sciences at age 15. One potential reason that has received little attention is that STEM programs are male-dominated, which may discourage female students from pursuing STEM in anticipation of the disadvantages of being a gender minority. In this paper, we examine whether the gender ratio affects students' college choices and whether the low female share in STEM leads to an inefficient allocation of talent, using an incentivized discrete choice experiment with high school students.

We find that the gender ratio affects the college choices of both female and male students. Both genders prefer programs with a balanced gender ratio over those dominated by either gender, and both prefer being a majority to being a minority. A decomposition reveals that students avoid being a minority primarily because they anticipate difficulty fitting in. Importantly, these preferences vary by context: female students with high mathematics ability are particularly averse to being a minority in STEM programs, while male students are generally less sensitive to the gender ratio in STEM than in non-STEM programs. A quantification exercise shows that the low female share in STEM, coupled with these preferences, generates inefficiency in talent allocation – high-mathematics-ability students are deterred from entering STEM, and their seats are filled by low-mathematics-ability students, with effects more pronounced among female students. Finally, using actual choice data collected several months later, we confirm that most students pursue fields consistent with their experimentally elicited preferences, but female students who prefer STEM are less likely to follow through – consistent with their stronger aversion to being a minority.

Our findings carry several implications. First, they suggest that the low female share in STEM is self-perpetuating: the current gender imbalance deters the very students who might otherwise enter, reinforcing the imbalance. This creates a coordination problem where individual students' rational responses to the existing composition collectively sustain an inefficient equilibrium. Second, our results provide an efficiency-based rationale for policies aimed at increasing female representation in STEM. Interventions that create more welcoming environments for female students can be justified not only on equity grounds but also because they may improve talent allocation.

Third, the finding that anticipated difficulty fitting in drives minority avoidance points toward specific mechanisms that interventions might target – for instance, providing information about peer support or highlighting successful integration of minority students. Such initiatives would benefit society beyond improving talent allocation: they can help address gender differences in occupational sorting and narrow the gender wage gap, shifting social norms toward greater gender equality in the long run. More broadly, more gender-equal societies benefit all members. In professional settings, they foster better recognition of ideas (Koffi 2025) and improve team performance (Hoogendoorn, Oosterbeek, and van Praag 2013). In social life, they reduce hostility toward same-sex marriage (Baranov, De Haas, and Grosjean 2023) and support higher rates of cross-gender friendships (Bailey et al. 2025).

Several limitations warrant discussion. Our sample consists of high-achieving students at selective high schools in the Greater Tokyo area, who face fewer constraints on their college choices than students elsewhere in Japan or in other countries. Whether our findings generalize to students with weaker mathematics ability or tighter geographic constraints remains an open question. Additionally, while our incentivized choice experiment provides clean identification of preferences, it cannot capture all factors that influence actual college decisions, including parental pressure, teacher recommendations, and peer influences.

Future research could extend this work in several directions. Examining whether preferences for gender balance vary across countries with different cultural norms around gender would help assess the generalizability of our findings. Investigating whether information interventions – such as highlighting successful female STEM students or providing realistic previews of the social environment—can attenuate minority avoidance would have direct policy relevance. Finally, tracking students longitudinally to examine whether those who enter gender-imbalanced programs experience the difficulties they anticipate would shed light on whether students' beliefs are accurately calibrated.

In conclusion, our findings suggest that the gender composition of STEM programs is not merely a symptom of other factors driving female students away, but is itself a barrier to entry. Breaking this cycle may require coordinated efforts to shift the equilibrium. Small increases in female representation could reduce the cost of entry for subsequent cohorts, potentially generating momentum toward greater balance. More broadly, our results underscore how compositional features of educational environments can shape who pursues which fields, with consequences for both individual careers and the efficient allocation of talent in society.

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Online Appendix

A Mixed Logit Model Performance

Table A1 presents mixed logit parameter estimates for female students (column 1) and male students (column 2). The table shows substantial and well-identified heterogeneity in preferences for both groups. The standard deviation of the random STEM coefficient is 1.62 for females and 1.60 for males, both highly statistically significant ($p < 0.001$). This implies wide variation in baseline STEM preferences. Heterogeneity in preferences for female share also exhibits substantial variation: for females, the standard deviations are approximately 0.28 for the linear term and 0.22 for the quadratic term, while for males they are approximately 0.37 and 0.42, respectively.

The model fit is adequate for both samples: the log-likelihood is -2,747.5 for females and -2,632.2 for males, and McFadden's pseudo-R-squared is 0.147 for females and 0.146 for males. These values confirm that the data contain sufficient within-individual variation to identify the distribution of random coefficients and the associated individual-level parameters.

The estimated covariance matrices display structured and interpretable correlation patterns, as shown in Table A2. For example, the correlation between the linear and quadratic female share terms is -0.364 for females and 0.704 for males, consistent with heterogeneous nonlinear responses. Moreover, the correlation between STEM times female share and female share is 0.876 for females and -0.834 for males, indicating systematic rather than idiosyncratic interaction heterogeneity.

B Additional Figures and Tables

Table A1: Mixed Logit Parameter Estimates

Sample:	Female	Male
Outcome:	Choose the program (0/1)	
	(1)	(2)
STEM	-0.449***	0.218***
	(0.075)	(0.074)
Female student share	0.144***	-0.065
	(0.039)	(0.040)
Female student share squared	-0.282***	-0.349***
	(0.043)	(0.045)
Selectivity index (population SD)	0.153***	0.267***
	(0.026)	(0.028)
Cohort size/100	0.039	0.052*
	(0.026)	(0.028)
Intl exchange program	0.370***	0.177**
	(0.066)	(0.070)
Club participation rate	0.139***	0.090***
	(0.026)	(0.028)
STEM x Female student share	0.008	-0.011
	(0.056)	(0.056)
STEM x Female student share squared	-0.046	0.015
	(0.063)	(0.061)
Intercept for right	-0.008	-0.056
	(0.037)	(0.038)
SD(STEM)	1.618***	1.595***
	(0.078)	(0.080)
Chol(FShare, FShare)	0.281***	0.365***
	(0.076)	(0.079)
Chol(FShare, FShare squared)	-0.079	0.178**
	(0.067)	(0.072)
Chol(FShare squared, FShare squared)	0.203**	0.180**
	(0.081)	(0.085)
Chol(FShare, FShare x STEM)	0.009	-0.111
	(0.111)	(0.110)
Chol(FShare, FShare squared x STEM)	0.013	-0.264***
	(0.080)	(0.084)
Chol(FShare squared, FShare x STEM)	0.120	-0.000
	(0.097)	(0.095)
Chol(FShare squared, FShare squared x STEM)	0.091	0.152
	(0.103)	(0.135)
Chol(FShare x STEM, FShare x STEM)	0.029	-0.171
	(0.488)	(0.166)
Chol(FShare x STEM, FShare squared x STEM)	-0.068	-0.078
	(0.714)	(0.154)
Chol(FShare squared x STEM, FShare squared x STEM)	0.015	-0.025
	(3.719)	(2.056)
Log Likelihood	-2747.457	-2632.201
McFadden's Pseudo-R-squared	0.147	0.146
No. observations	9298	8902
No. students	310	297

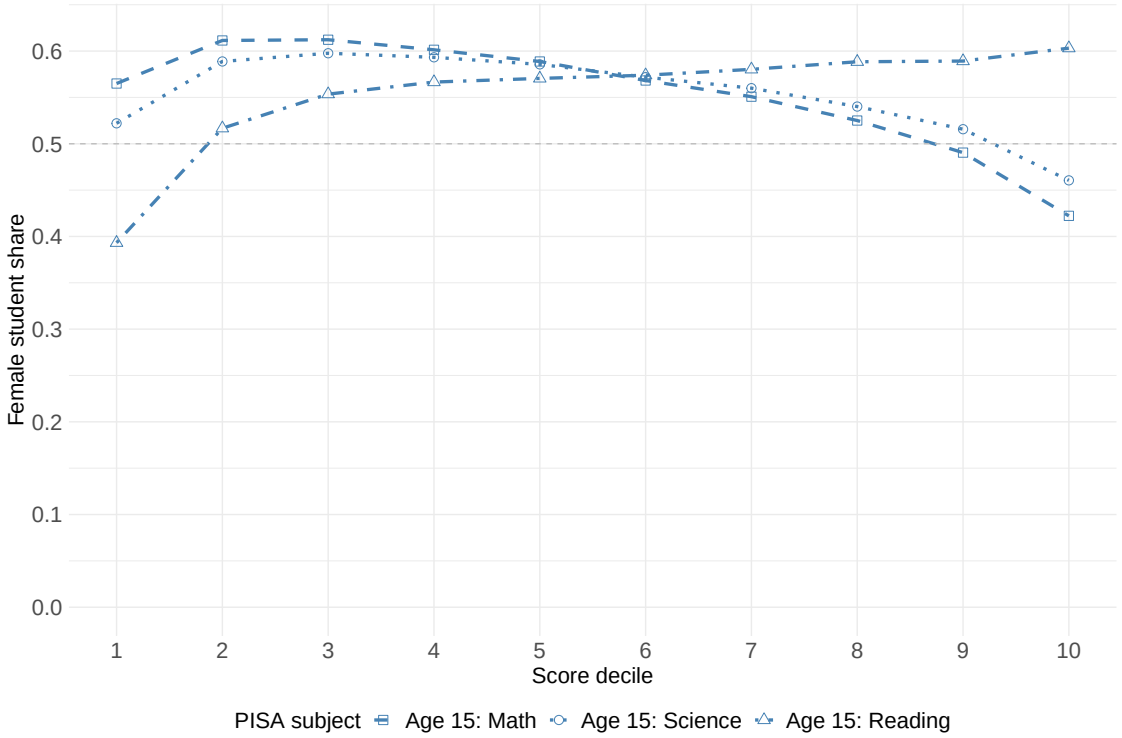
Notes: This table presents mixed logit parameter estimates for female students (column 1) and male students (column 2). Significance levels: * 10%, ** 5%, and *** 1%.

Table A2: Correlation Matrix of Mixed Logit Individual-Level Parameters

Panel A: Females				
	Female share	Female share squared	STEM x Female share	STEM x Female share squared
Female share	1.000	-0.364	0.074	0.117
Female share squared	-0.364	1.000	0.876	0.692
STEM x Female share	0.074	0.876	1.000	0.636
STEM x Female share squared	0.117	0.692	0.636	1.000
Panel B: Males				
	Female share	Female share squared	STEM x Female share	STEM x Female share squared
Female share	1.000	0.704	-0.543	-0.836
Female share squared	0.704	1.000	-0.384	-0.247
STEM x Female share	-0.543	-0.384	1.000	0.663
STEM x Female share squared	-0.836	-0.247	0.663	1.000

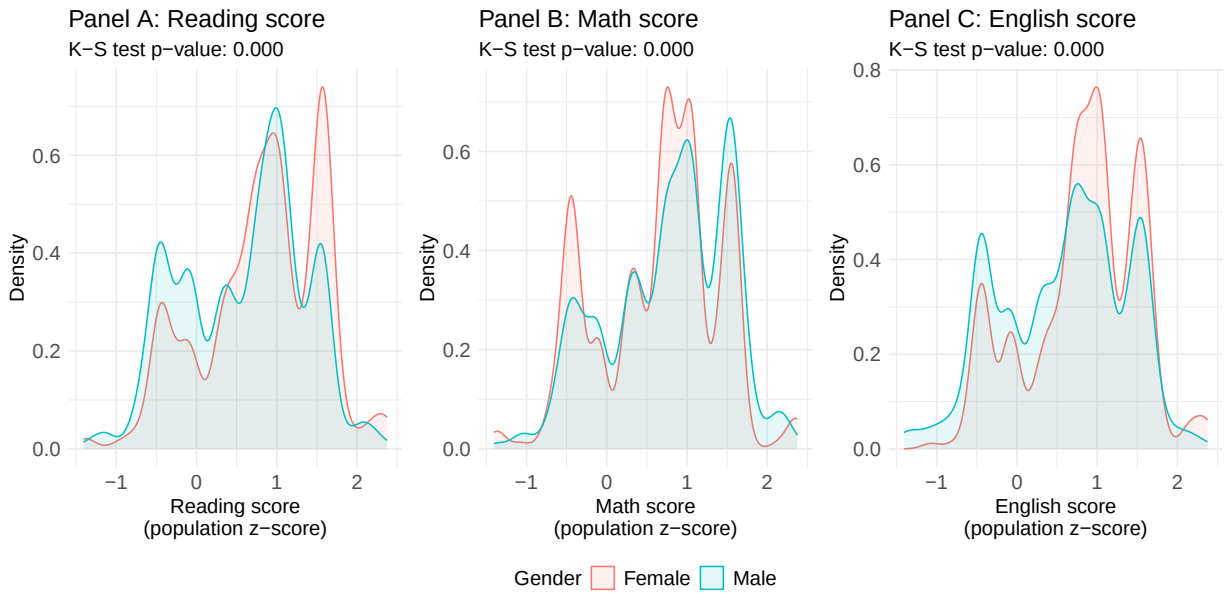
Notes: This table reports the correlation matrix of mixed logit individual-level parameters for female (Panel A) and male (Panel B) students.

Figure B1: Female share at age 15 by subject – OECD Average



Notes: This figure plots the OECD average share of female students in each of the 10 deciles of the PISA mathematics (dark dashed), science (dark dotted), and reading (dark dash-dotted) score distributions among 15 year-old students in 2018 who plan to pursue tertiary education. *Sources:* OECD (2018).

Figure B2: Distribution of Abilities



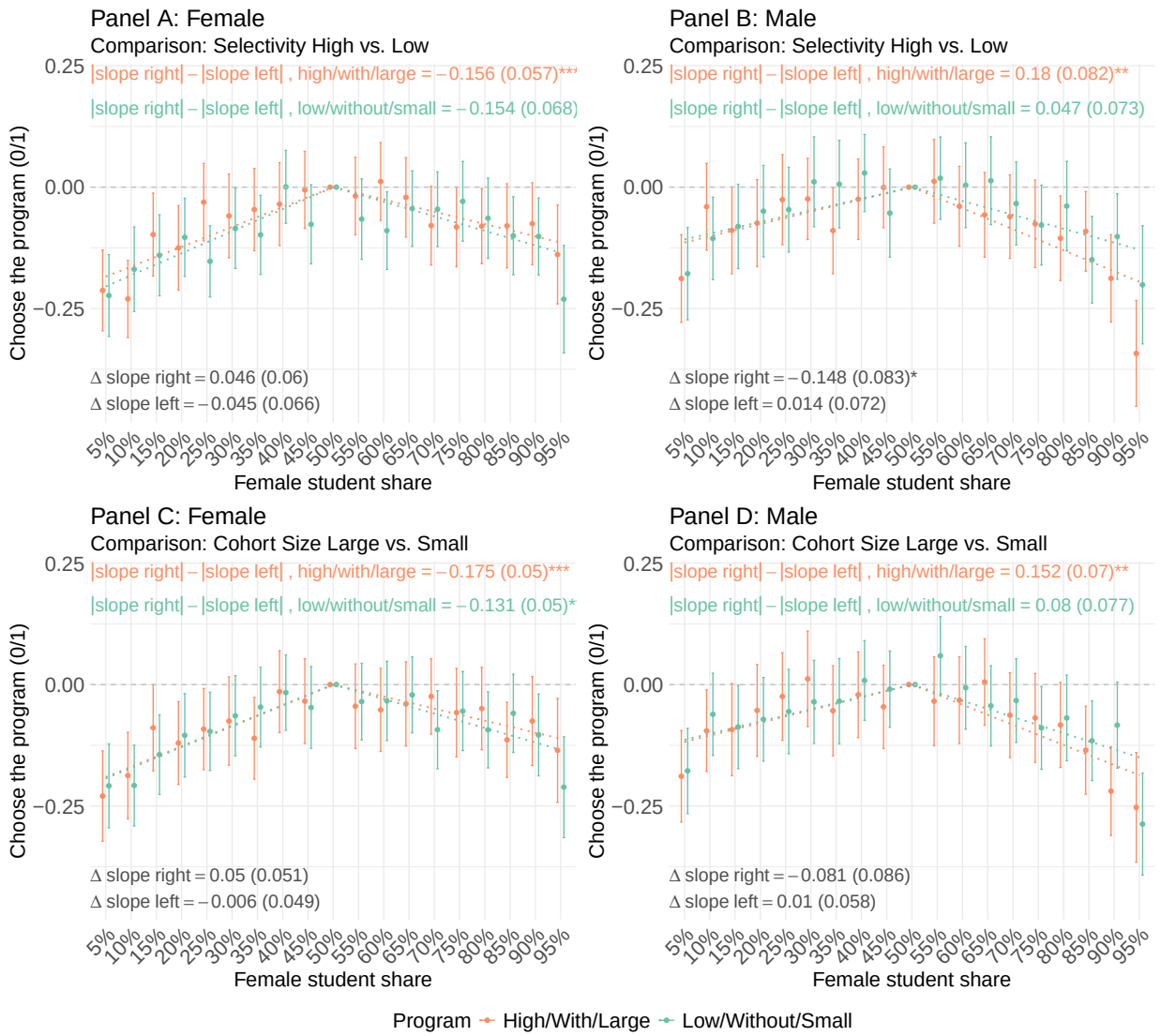
Notes: This figure presents the distribution of abilities of female and male students in reading (Panel A), mathematics (Panel B), and English (Panel C). K-S test p-value shows the Kolmogorov-Smirnov test p-value for differences in the distribution between female and male students.

Table B3: Attribute Values

General attributes	
College name:	AA, AB, AC, AD, AE, AF, AG, AH, AI, AJ, AK, AL, AM, AN, AO, AP, AQ, AR, AS, AT, AU, AV, AW, AX, AY, AZ, BA, BB, BC, BD
Department:	
Non-STEM:	Literature, Law, Business, Economics, Sociology, Foreign Language
STEM:	Physics, Chemistry, Biology, Engineering, Information Technology, Agriculture
Department attributes	
Selectivity index:	55, 57.5, 60, 62.5, 65, 67.5, 70, 72.5
Cohort size:	200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900
Female student share:	5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%
College attributes	
International exchange program:	Yes, Yes, Yes, Yes, No
Club participation rate:	40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%

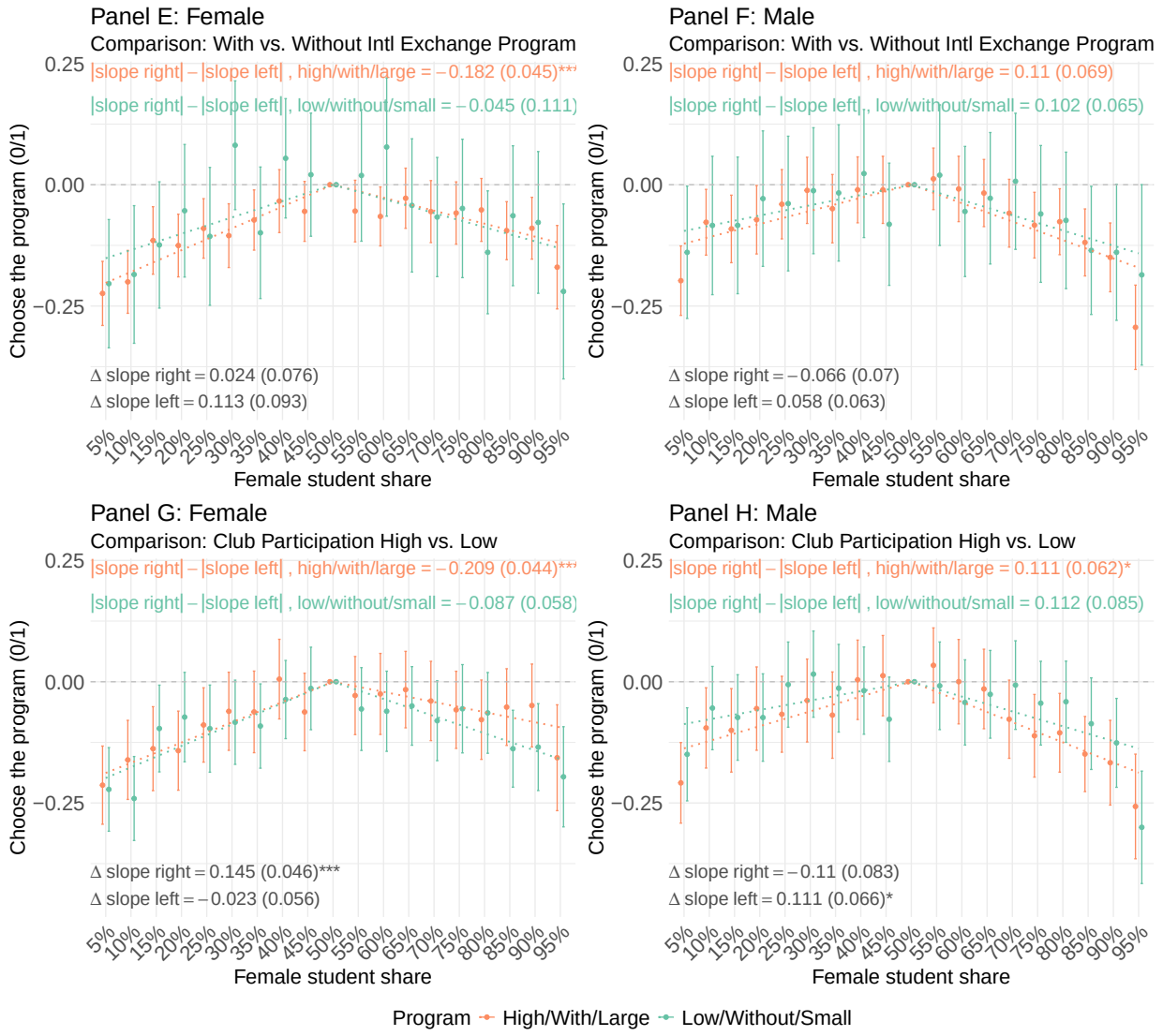
Notes: This table presents the values each attribute can take in a given program.

Figure B3: Heterogeneity of Preferences for the Gender Ratio by Selectivity and Cohort Size



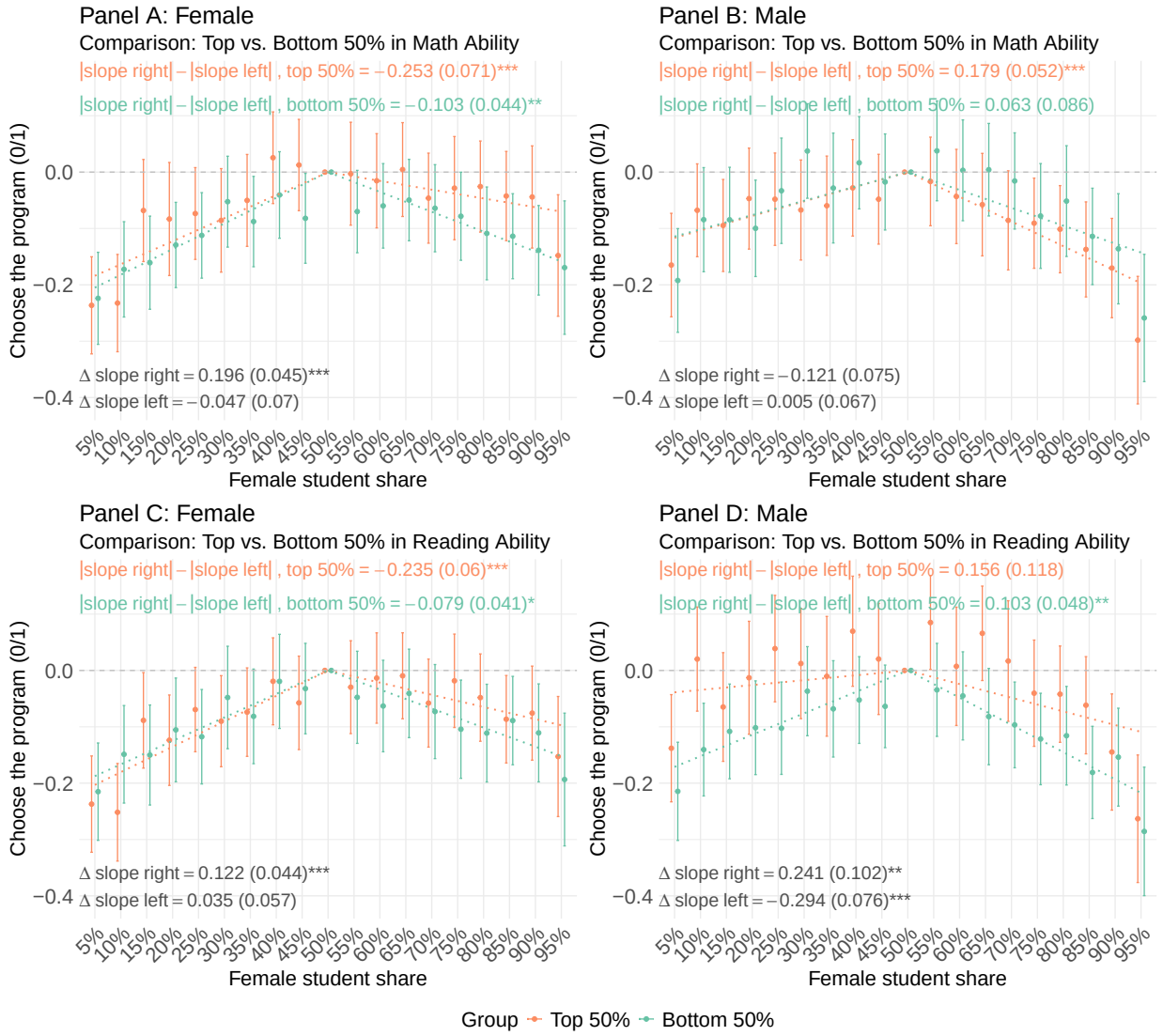
Notes: This figure plots coefficient estimates and the 95% confidence intervals for female and male students where we interact the attributes with an indicator variable for program attributes other than STEM vs. non-STEM. Panels A and B interact with an indicator variable for high selectivity program and Panels C and D interact with an indicator variable for large cohort size program. Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for each side (below 50% and above 50%) for two groups. We imposed the constraint that the lines pass through the 50% point. The differences in the slopes between the two groups are calculated using the weighted least squares standard errors. Significance levels: * 10%, ** 5%, and *** 1%.

Figure B4: Heterogeneity of Preferences for the Gender Ratio by Intl Exchange and Club Participation



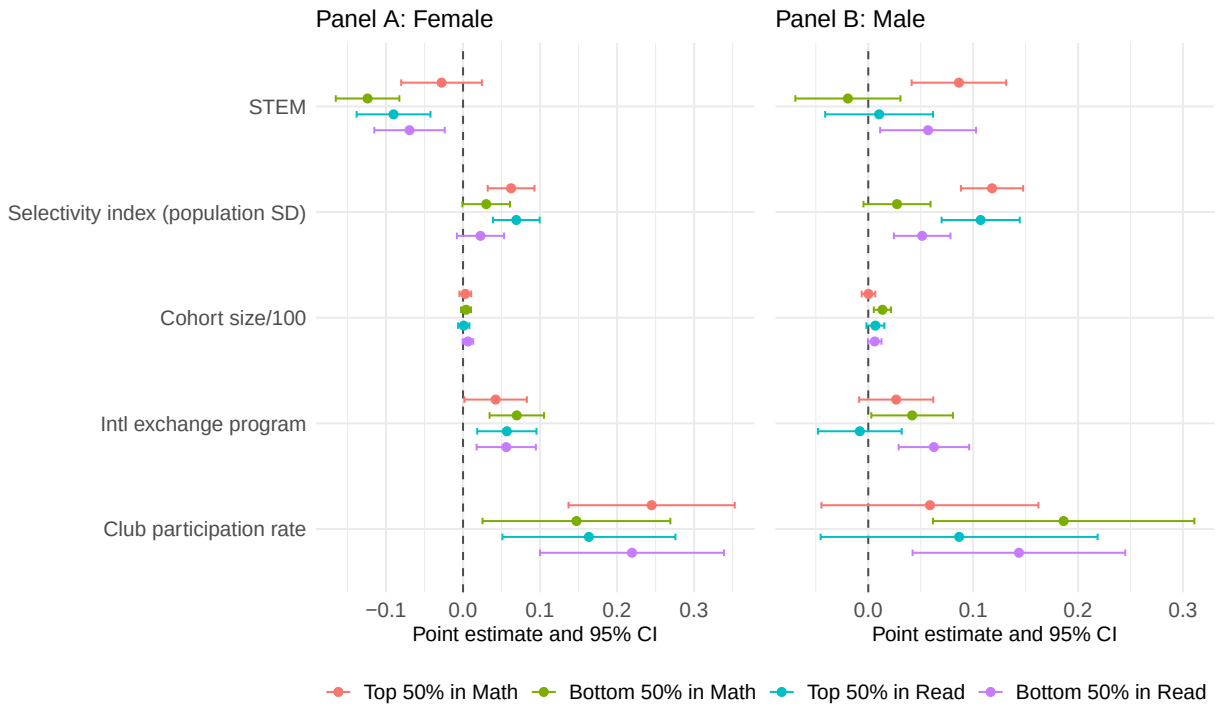
Notes: This figure plots coefficient estimates and the 95% confidence intervals for female and male students where we interact the attributes with an indicator variable for program attributes other than STEM vs. non-STEM. Panels A and B interact with an indicator variable for programs with international exchange program and Panels C and D interact with an indicator variable for high club participation rate. Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for each side (below 50% and above 50%) for two groups. We imposed the constraint that the lines pass through the 50% point. The differences in the slopes between the two groups are calculated using the weighted least squares standard errors. Significance levels: * 10%, ** 5%, and *** 1%.

Figure B5: Heterogeneity of Preferences for the Gender Ratio by Abilities



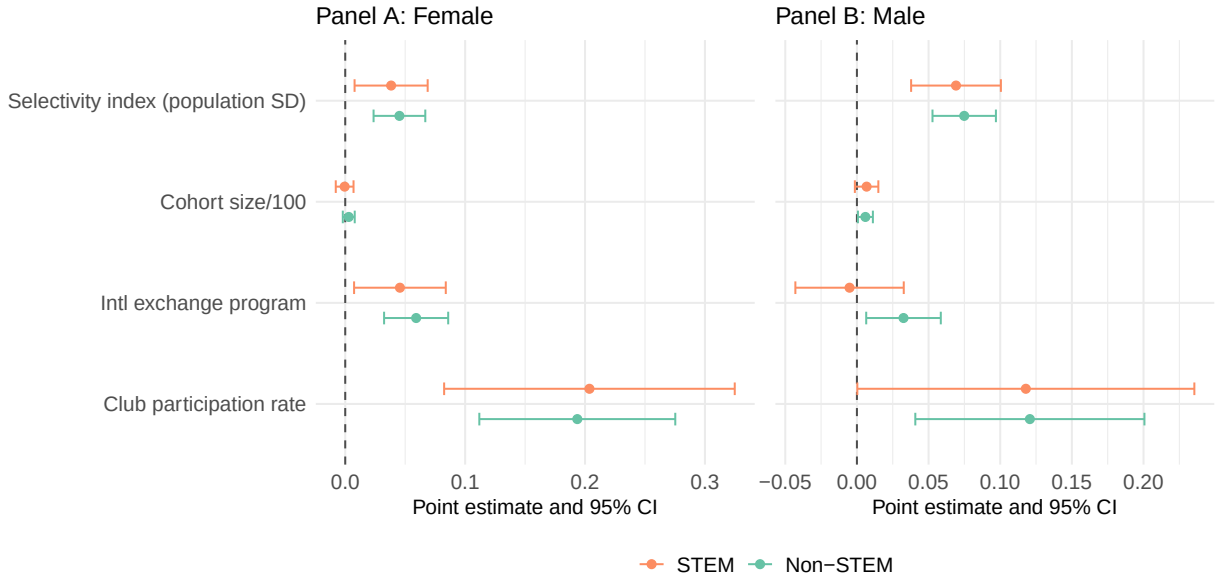
Notes: This figure plots coefficient estimates and the 95% confidence intervals for female and male students where we interact the attributes with an indicator variable for academic abilities. Panels A and B interact with an indicator variable for top 50% mathematics abilities students and Panels C and D interact with an indicator variable for top 50% reading abilities students. Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for each side (below 50% and above 50%) for two groups. We imposed the constraint that the lines pass through the 50% point. The differences in the slopes between the two groups are calculated using the weighted least squares standard errors. Significance levels: * 10%, ** 5%, and *** 1%.

Figure B6: Heterogeneity of Preferences for Other Attributes by Ability



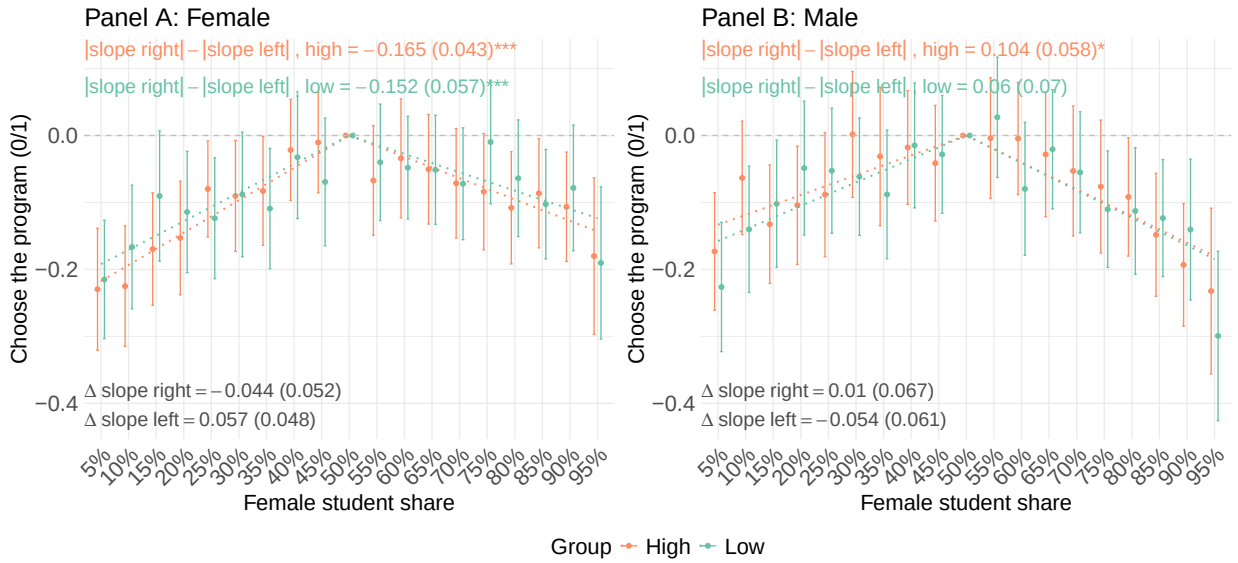
Notes: This figure plots coefficient estimates and the 95% confidence intervals of program attributes for female (Panel A) and male students (Panel B) with different academic abilities. Female student share is included in the estimation but omitted from this figure for brevity. Standard errors are clustered at the student level.

Figure B7: Heterogeneity of Preferences for Other Attributes – STEM vs. Non-STEM



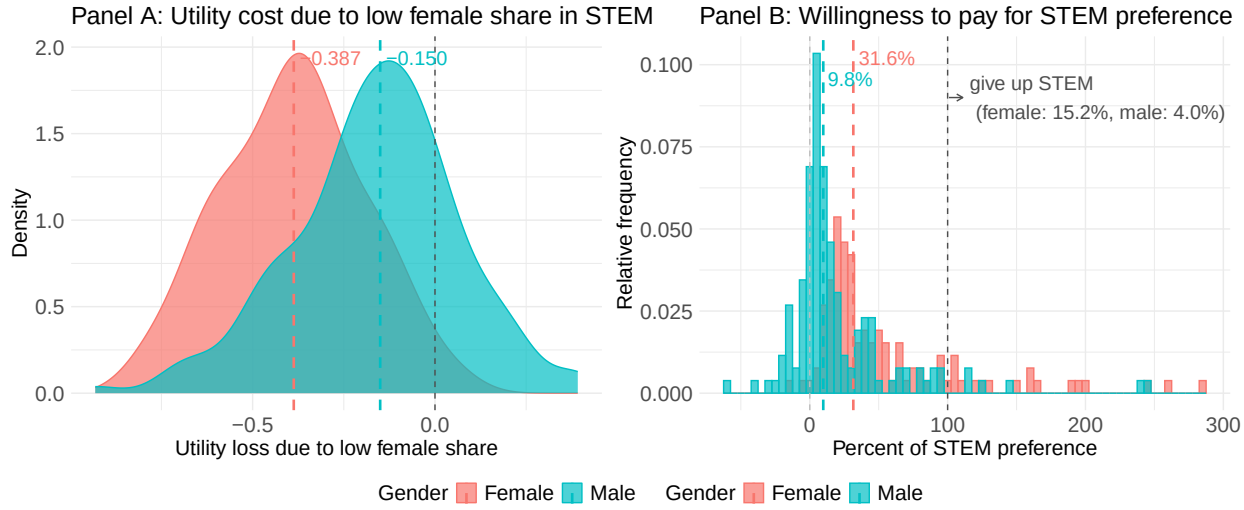
Notes: This figure plots coefficient estimates and the 95% confidence intervals of program attributes for female (Panel A) and male students (Panel B) for STEM and non-STEM programs. Female student share is included in the estimation but omitted from this figure for brevity. Standard errors are clustered at the student level.

Figure B8: Heterogeneity of the Preferences by Parental Education Levels



Notes: This figure plots coefficient estimates and the 95% confidence intervals for female and male students, just like Figure 5, but interact the attributes with an indicator variable for whether both parents have bachelor's degrees or above (High if 1, Low if 0). Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for each side (below 50% and above 50%) for two groups. We imposed the constraint that the lines pass through the 50% point. The differences in the slopes between High and Low are calculated using the weighted least squares standard errors. Significance levels: * 10%, ** 5%, and *** 1%.

Figure B9: Utility Cost and Willingness to Pay



Notes: This figure is restricted to students who prefer STEM. Panel A shows the utility cost due to the low female share in STEM. Panel B shows students' willingness to pay for an increase in the female share, expressed as a fraction of their STEM preference; we exclude the bottom 10% of students by STEM preference, as weak preferences inflate willingness-to-pay estimates.

Figure B10: Allocation Across Programs with the Actual Female Share

Notes: This figure plots student allocations across programs in the Greater Tokyo area under actual female shares, with each dot represents a program. Orange dots are STEM programs and green dots are non-STEM programs. We use the same student-proposing deferred acceptance algorithm as in Figure 8 with a few minor alternations to take account for these programs' actual capacity. Due to the small capacity in each program, we draw 2,000 independent shocks from a Type I extreme value distribution for each student's utility, compute 2,000 allocations, and use the average as each student's allocation.

Table B4: Preferences for Program Attributes – Reasons as Dependent Variables

Sample:	Female	Male	Female	Male	Female	Male	Female	Male
Outcome:	Fit in		Role model		Studies		Career	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
STEM	-0.037***	0.007	-0.047***	0.011	-0.061***	0.041***	-0.062***	0.021
	(0.012)	(0.013)	(0.013)	(0.013)	(0.015)	(0.015)	(0.016)	(0.017)
Female student share	1.819***	1.099***	0.708***	0.605***	0.239***	0.453***	0.336***	0.407***
	(0.089)	(0.092)	(0.085)	(0.089)	(0.086)	(0.084)	(0.087)	(0.085)
Female student share squared	-1.477***	-1.384***	-0.621***	-0.685***	-0.225***	-0.503***	-0.320***	-0.450***
	(0.094)	(0.093)	(0.084)	(0.089)	(0.084)	(0.086)	(0.086)	(0.086)
Selectivity index (population SD)	-0.014	0.015	0.088***	0.080***	0.084***	0.094***	0.075***	0.080***
	(0.009)	(0.010)	(0.011)	(0.011)	(0.015)	(0.014)	(0.010)	(0.011)
Cohort size/100	0.002	0.005*	0.012***	0.013***	-0.002	0.000	0.004*	0.006**
	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Intl exchange program	0.021	0.020	0.049***	0.040***	0.034***	0.047***	0.053***	0.051***
	(0.014)	(0.013)	(0.014)	(0.012)	(0.013)	(0.013)	(0.014)	(0.015)
Club participation rate	0.282***	0.200***	0.651***	0.443***	0.086**	0.070*	0.165***	0.158***
	(0.042)	(0.043)	(0.054)	(0.050)	(0.039)	(0.041)	(0.040)	(0.043)
Constant	0.504***	0.479***	0.518***	0.490***	0.483***	0.480***	0.494***	0.495***
	(0.007)	(0.007)	(0.007)	(0.008)	(0.007)	(0.008)	(0.007)	(0.008)
Adj. R-squared	0.159	0.100	0.104	0.062	0.032	0.037	0.035	0.029
No. observations	4649	4451	4649	4451	4649	4451	4649	4451
No. students	310	297	310	297	310	297	310	297

Notes: This table presents the same specifications as Table 2 but with indicator variables for the four reasons in place of choice as the outcome variables. Significance levels: * 10%, ** 5%, and *** 1%.

Table B5: Preferences for Program Attributes – Logit

Sample:	Female		Male		All	
Outcome:	Choose the program (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
STEM	-0.078***	-0.078***	0.038**	0.038**	0.037**	0.038**
	(0.018)	(0.018)	(0.018)	(0.017)	(0.017)	(0.017)
Female student share	0.090***	0.819***	-0.036	0.815***	-0.036	0.811***
	(0.022)	(0.097)	(0.024)	(0.097)	(0.024)	(0.094)
Female student share squared		-0.742***		-0.872***		-0.868***
		(0.095)		(0.099)		(0.096)
Selectivity index (population SD)	0.046***	0.046***	0.074***	0.074***	0.074***	0.074***
	(0.011)	(0.011)	(0.012)	(0.012)	(0.012)	(0.012)
Cohort size/100	0.003	0.003	0.006**	0.006**	0.006**	0.006**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Intl exchange program	0.059***	0.060***	0.033**	0.034**	0.032**	0.034**
	(0.014)	(0.014)	(0.013)	(0.013)	(0.013)	(0.013)
Club participation rate	0.193***	0.196***	0.122***	0.130***	0.121***	0.130***
	(0.042)	(0.043)	(0.041)	(0.041)	(0.041)	(0.041)
Female					0.016	0.015
					(0.011)	(0.011)
STEM x Female					-0.116***	-0.116***
					(0.025)	(0.025)
Female student share x Female					0.127***	0.011
					(0.033)	(0.131)
Female student share squared x Female						0.123
						(0.131)
Selectivity index (population SD) x Female					-0.028*	-0.027*
					(0.016)	(0.016)
Cohort size/100 x Female					-0.003	-0.003
					(0.004)	(0.004)
Intl exchange program x Female					0.027	0.026
					(0.019)	(0.019)
Club participation rate x Female					0.074	0.067
					(0.059)	(0.059)
Log Likelihood	-3133.13	-3090.73	-3032.28	-2977.35	-6165.41	-6068.08
No. observations	4649	4649	4451	4451	9100	9100
No. students	310	310	297	297	607	607

Notes: This table presents the logit coefficient estimates in average marginal effects on the program attributes with choice as the dependent variable. Columns 1 and 2 present estimates for female students, columns 3 and 4 present estimates for male students, and columns 5 and 6 present estimates for differences between female and male students. The average marginal effects for constant term are undefined and thus are not shown. Standard errors are clustered at the student level. Significance levels: * 10%, ** 5%, and *** 1%.

C Post-Experimental Questionnaire

Post-Experimental Questionnaire (English translation)

Questionnaire 1/4

Please tell us about yourself and your family.

- Your gender: [Male, Female, Non-binary or Other]
- Your father's academic background: [Below high school, High school, Vocational school, Associate degree, Bachelor's degree, Master's degree or above, I do not know]
- Your mother's academic background: [Below high school, High school, Vocational school, Associate degree, Bachelor's degree, Master's degree or above, I do not know]
- Extra schooling per week: [No extra schooling, one day a week, two days a week, three days a week, four days a week, five days a week or more]

Please recall the exam held on [Month Day]. What was your score in the following subjects?

- Reading: [Integer]
- Mathematics: [Integer]
- English: [Integer]

Please click "→" to proceed.

——————————— Page break ———————————

Questionnaire 2/4

Do you think **your scores in the exam held on [Month Day] accurately reflect your abilities?** Please answer for each of the subjects below.

	My abilities are lower than the score	My abilities are slightly lower than the score	It reflects my ability accurately	My abilities are slightly higher than the score	My abilities are higher than the score
Reading	○	○	○	○	○
Mathematics	○	○	○	○	○
English	○	○	○	○	○

- Do you consider yourself someone who is **willing to compete with others**, or someone who **avoids competing with others**? [Avoid competing with others, Slightly avoid competing with others, Neither avoid nor willing to compete with others, Slightly willing to compete with others, Willing to compete with others]

- Do you consider yourself someone who is generally **willing to take risks**, or someone who **avoids taking risks**? [Avoid taking risks, Slightly avoid taking risks, Neither avoid nor willing to take risks, Slightly willing to take risks, Willing to take risks]

Please click "→" to proceed.

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Questionnaire 3/4

What do you think is **the average female-student ratio** in the following departments across the colleges in Japan?

	Below 10%	11-20%	21-30%	31-40%	41-50%	51-60%	61-70%	71-80%	81-90%	91% or above
Humanities Departments (Literature, history, philosophy, etc.)	○	○	○	○	○	○	○	○	○	○
Social Sciences Departments (Law, Economics, Sociology, etc.)	○	○	○	○	○	○	○	○	○	○
Sciences and Engineering Departments (Physics, Biology, Mechanical Engineering, Information Technology, etc.)	○	○	○	○	○	○	○	○	○	○
Medicine and Nursing Departments (Medicine, Dentistry, Pharmacy, Nursing, etc.)	○	○	○	○	○	○	○	○	○	○

Please click "→" to proceed.

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Questionnaire 4/4

Please tell us your opinion about this survey.

- Was it easy to follow? [Difficult to follow, Slightly difficult to follow, Neither difficult nor easy to follow, Slightly easy to follow, Easy to follow]
- Which parts did you find it difficult to answer? [Text]
- What do you think is the purpose of this survey? [Text]
- Other comments? (optional) [Text]